



The Electromagnetic Performance of the Dual Readout Calorimeter in 2024 Test Beam Experiment

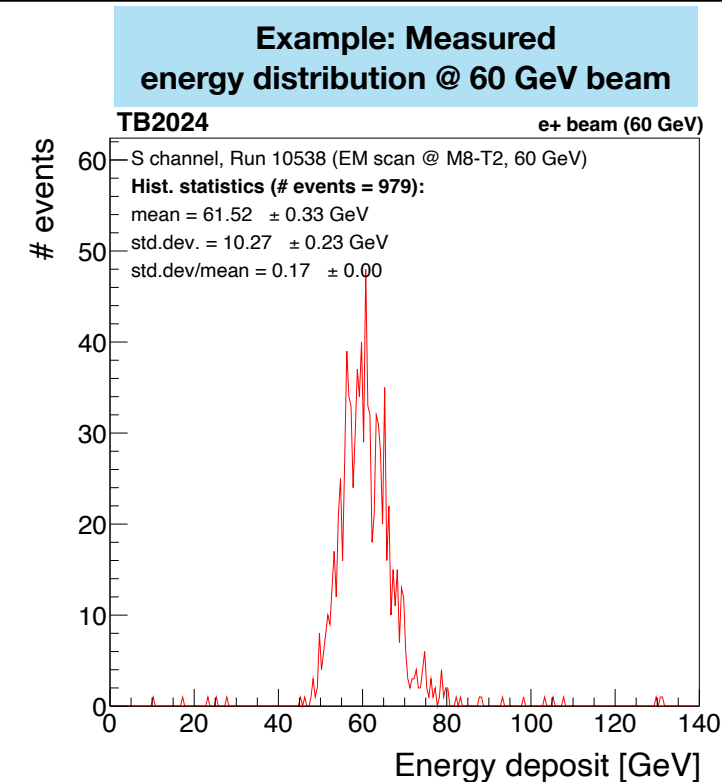
Kyeongpil Lee
Yonsei University

on behalf of Korea Dual Readout Calorimeter Collaboration

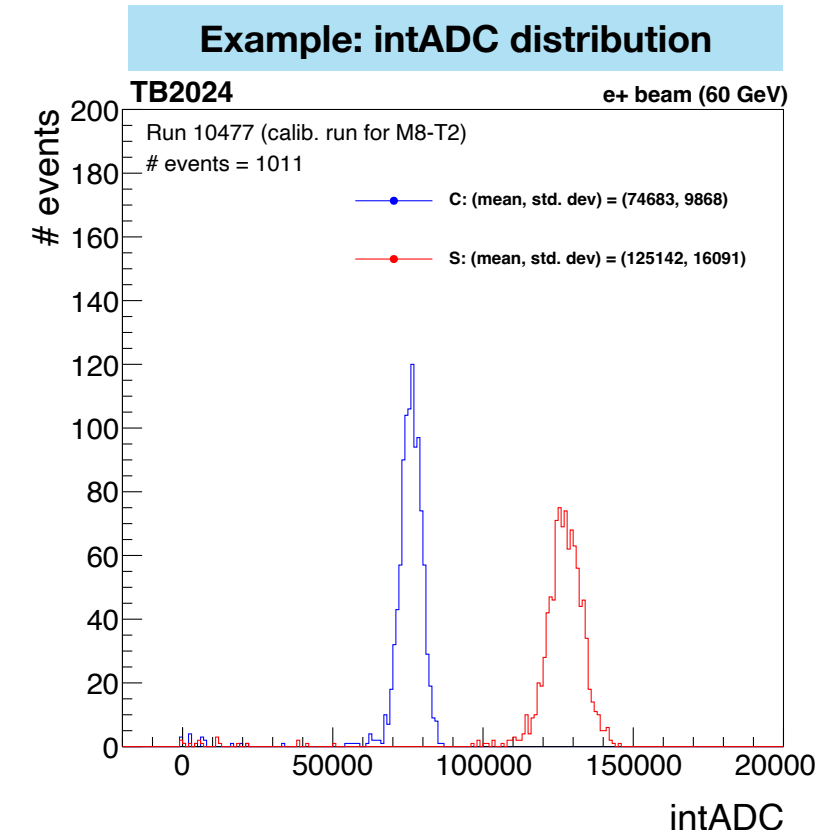
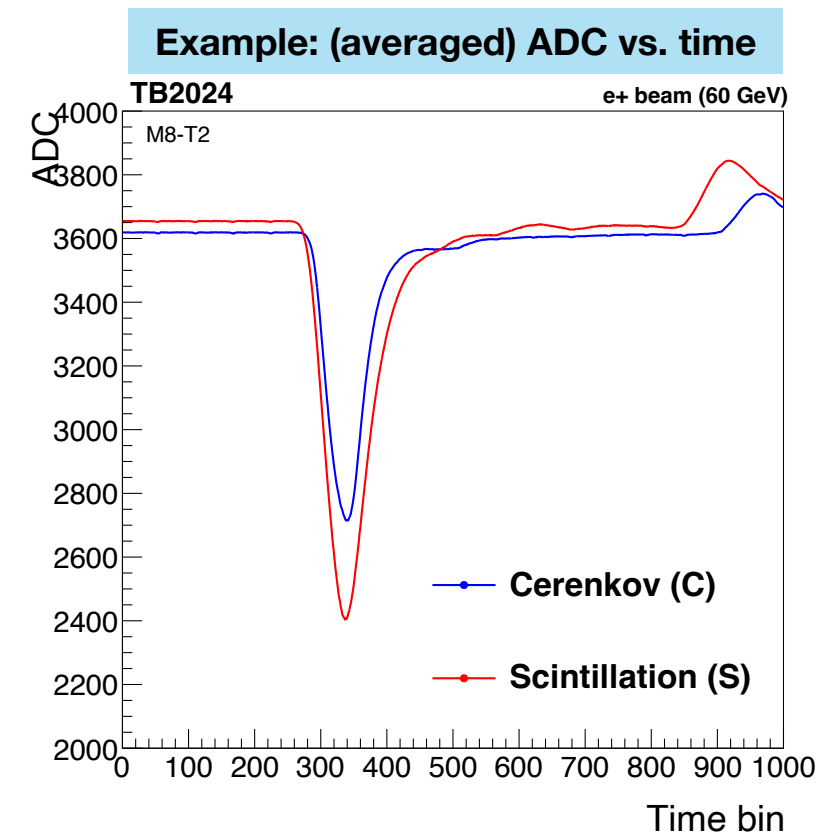
2024 KPS Fall Meeting ([session](#))
22-26 October, 2024

Introduction: Electromagnetic Performance

- ◉ Electromagnetic (EM) performance for a given beam energy: mainly two quantities for energetic **electrons & photons**
 - [1] **Scale** ($\langle E_{\text{measured}} \rangle / E_{\text{beam}}$):
How **close** $\langle \text{measured energy} \rangle$ is to the beam energy
 - [2] **Resolution**:
how well **focused** the measured energy is from its mean value
 - Usually **represented by** σ/E from the (gaussian) fit on the measured energy distribution
- ◉ The EM performance of Dual Readout Calorimeter (DRC) is studied **for different energies**
 - On the **seven** energy points with **e+** beam: 10, 20, 30, 40, 60, 80, 100 GeV
 - [1] **Scale vs. E ("linearity")**: How much the scale is stable over the different energies
 - [2] **Resolution vs. E**: How the resolution is changed over the energy & is the distribution following the expected shape
 - Expected to show better resolution in higher energy following $1/\sqrt{E}$ scaling



- ◉ [1] Determination of **integration range** for the DRC signal
 - Integral of ADC over a range ("intADC")
 $\propto$ energy deposit in the tower
 - Take the range covering the DRC signal (peak) in ADC vs. time distribution
 - Based on averaged shape over many events ("averaged time structure")
- ◉ [2] **Event selection**
 - To keep the events from the pure e^+ beam only
 - Use the signal from auxiliary detectors
 - Delayed wire chamber (DWC), Preshower (PS) and Muon counter (MC)
- ◉ [3] **Calibration: convert "intADC" to energy deposit [GeV]**
 - Using the data with 60 GeV e^+ beam for each tower
 - Multiply a constant to convert the obtained intADC to a energy unit
- ◉ [4] **EM performance results**
 - Obtain the scale & resolution per each energy point



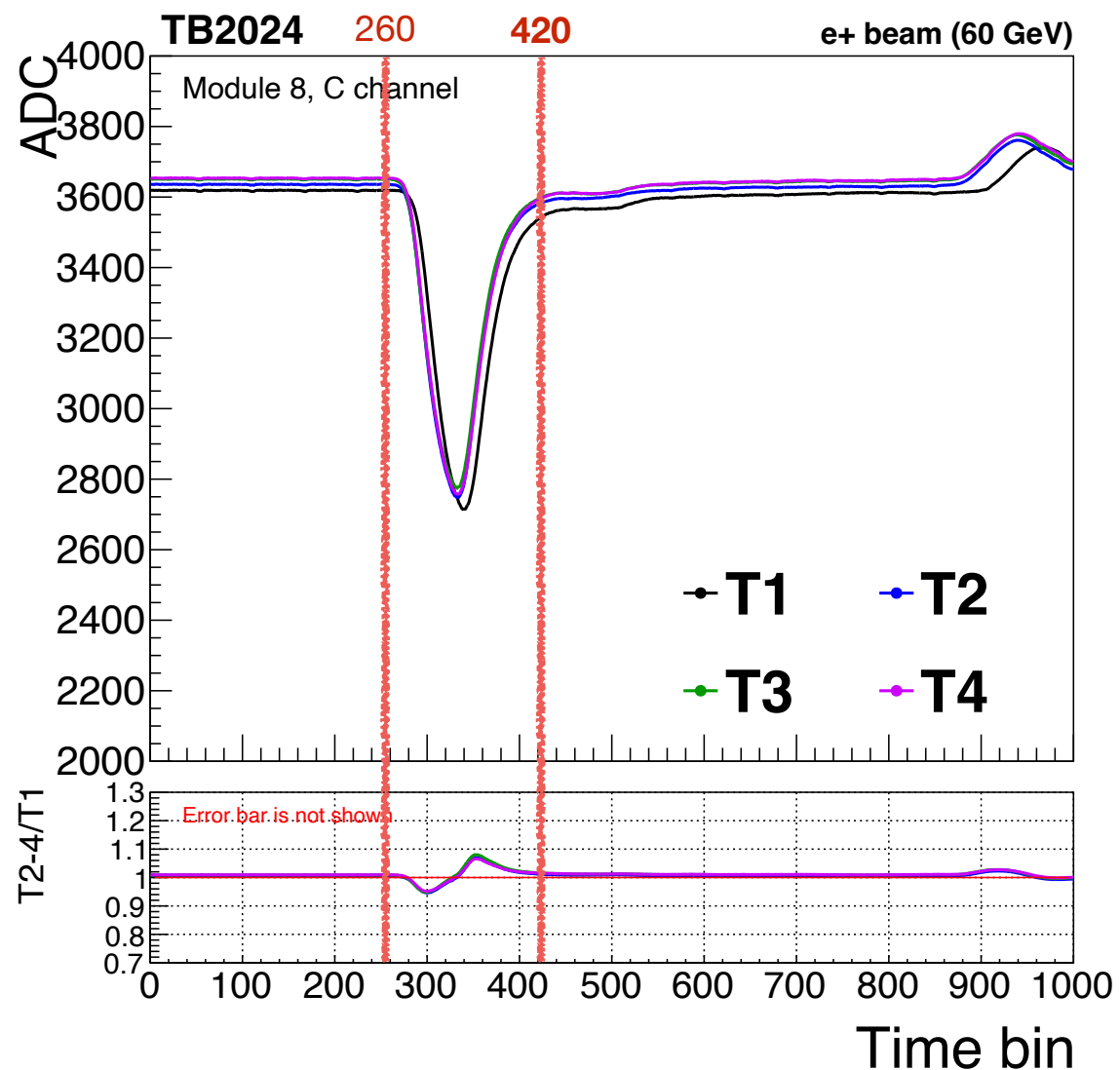
- Determine the range that covers the DRC signal (peak)

- Using the averaged ADC vs. time ("averaged time structure")
- S channel has larger signal than C channel: take a larger range to fully cover the peak
- [260, 420] for C channel; [260, 460] for S channel**
- The range covers the peak well over all modules
 - Except for M5 due to its (slightly) different timing: dedicated range is applied (backup)

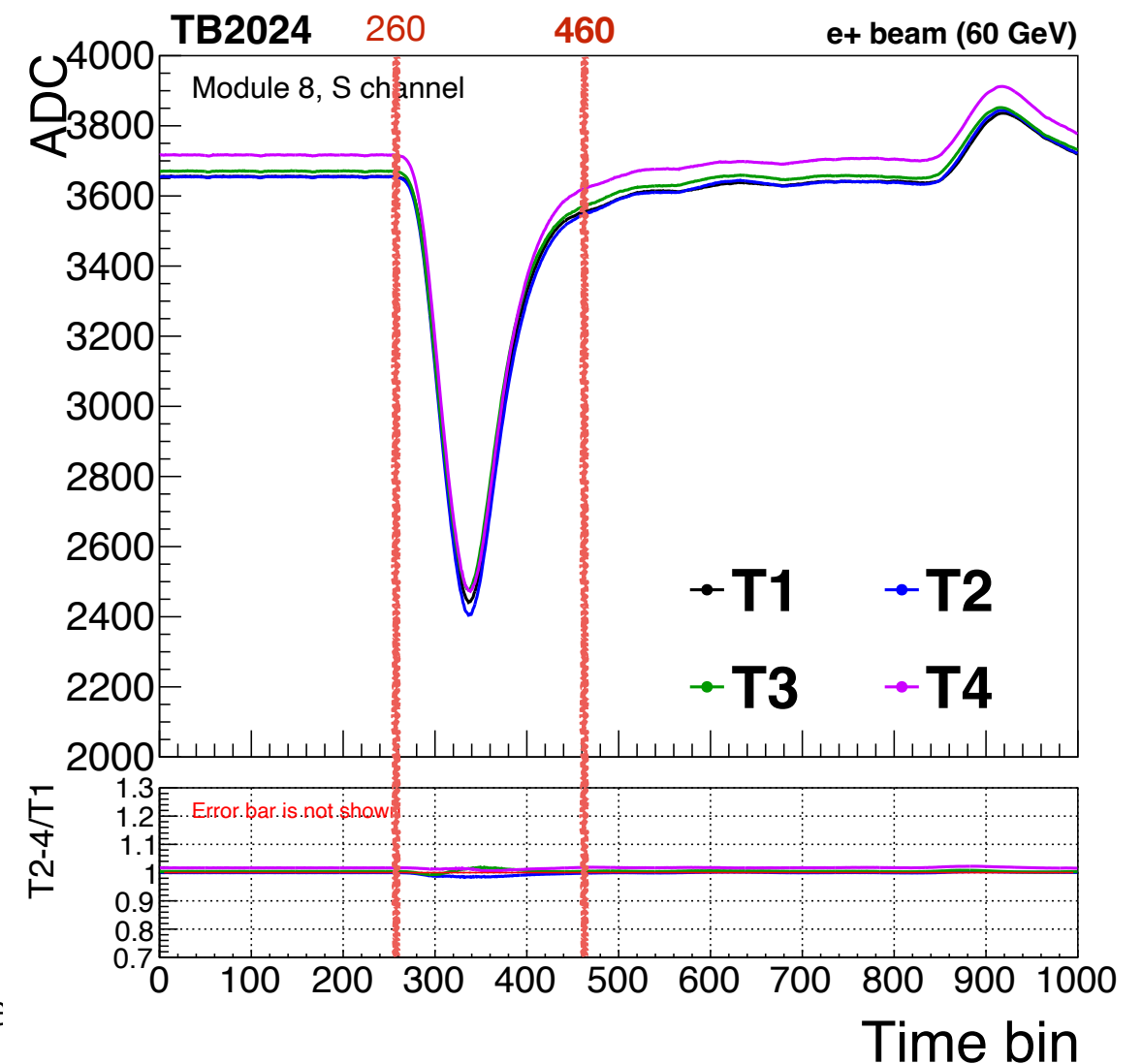
M1			
T1	T2	M2	M3
T3	T4		
M4		M5	M6
M7	M8	M9	

Example of an average time structure: Module 8

Cherenkov (C) channel



Scintillation (S) channel



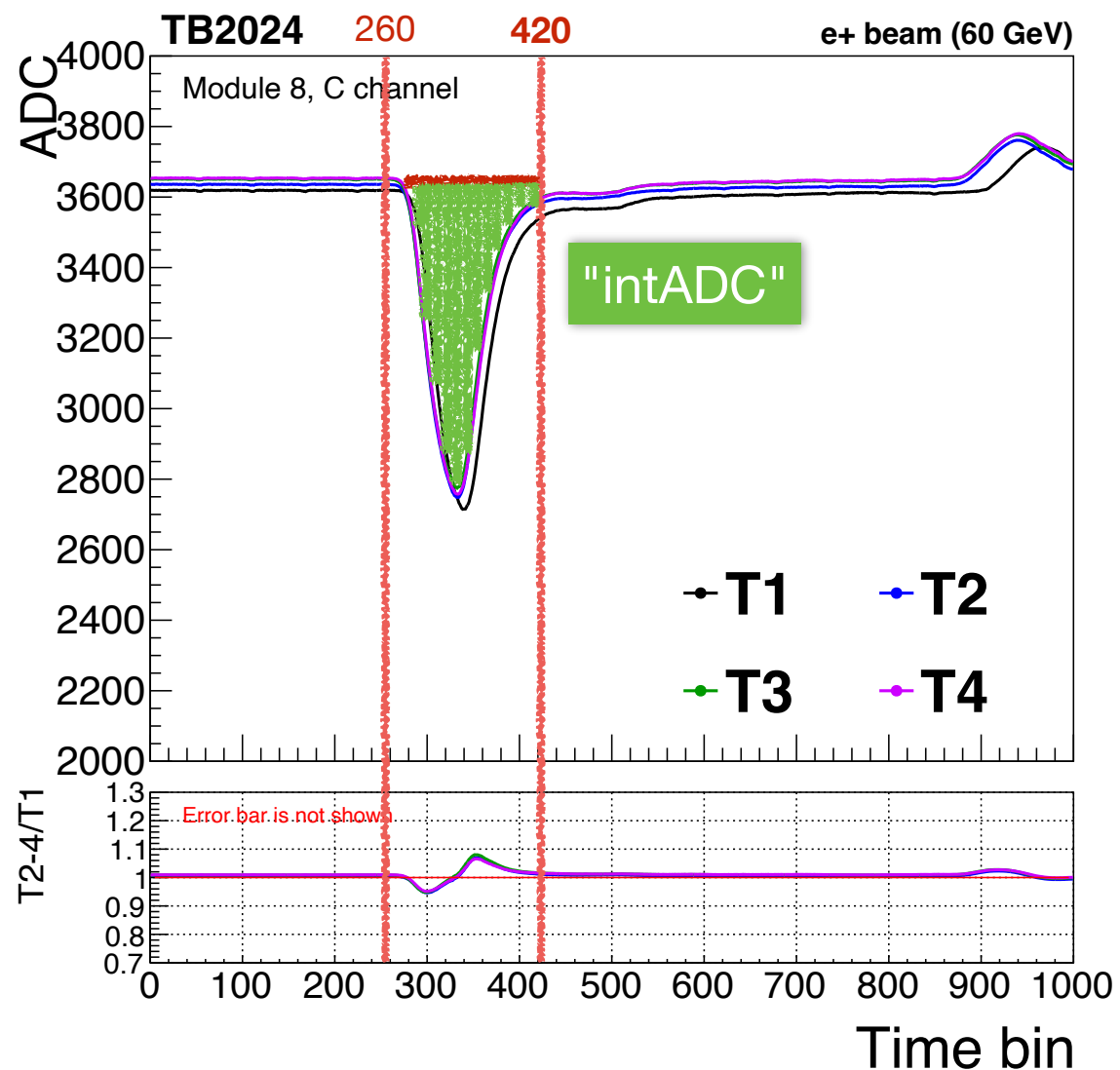
- Determine the range that covers the DRC signal (peak)

- Using the averaged ADC vs. time ("averaged time structure")
- S channel has larger signal than C channel: take a larger range to fully cover the peak
- [260, 420]** for C channel; **[260, 460]** for S channel
- Integral of ADC:** sum of all ADCs within the range from the pedestal

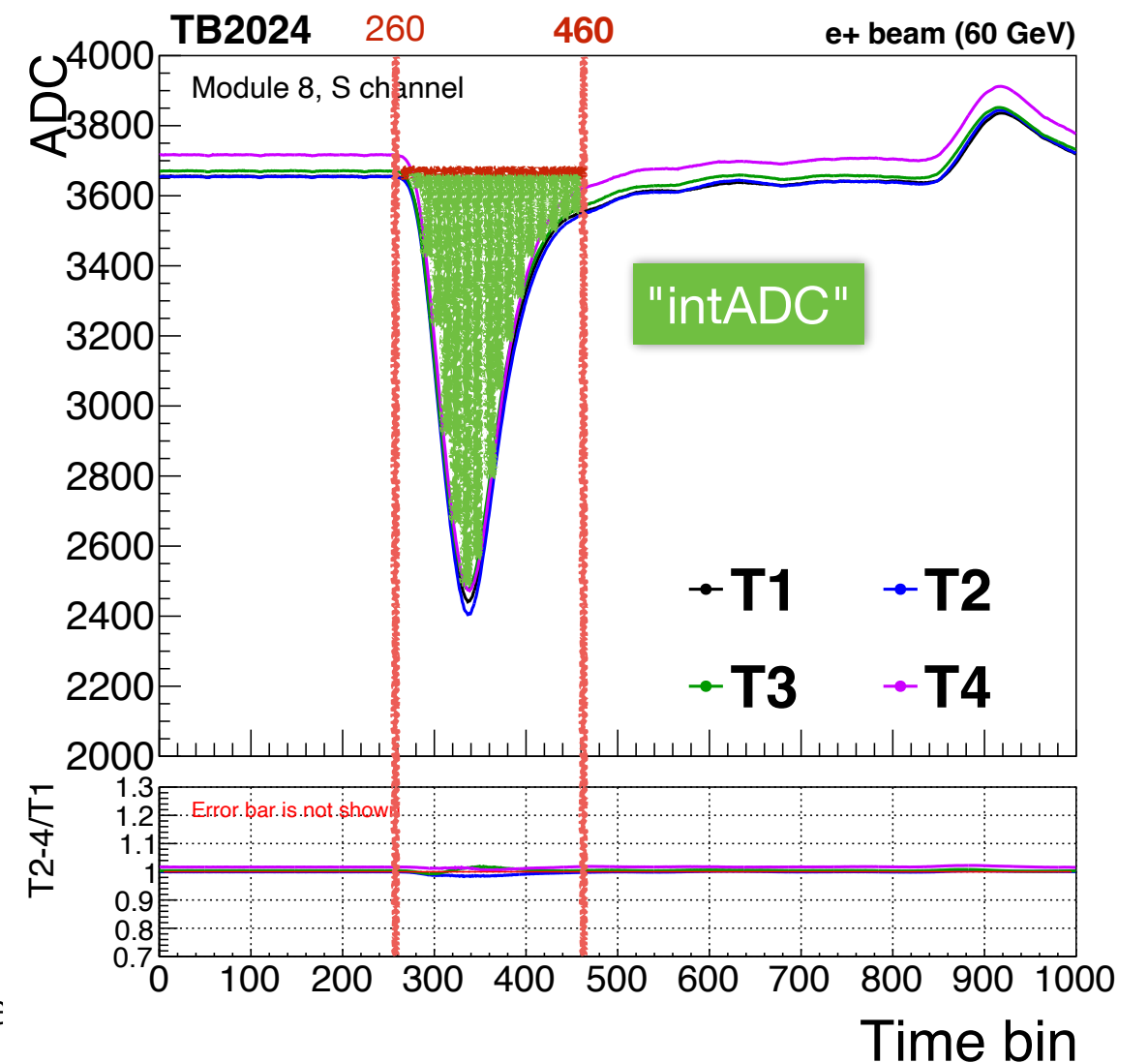
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T1	T2	M2	M3
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Example of an average time structure: Module 8

Cherenkov (C) channel



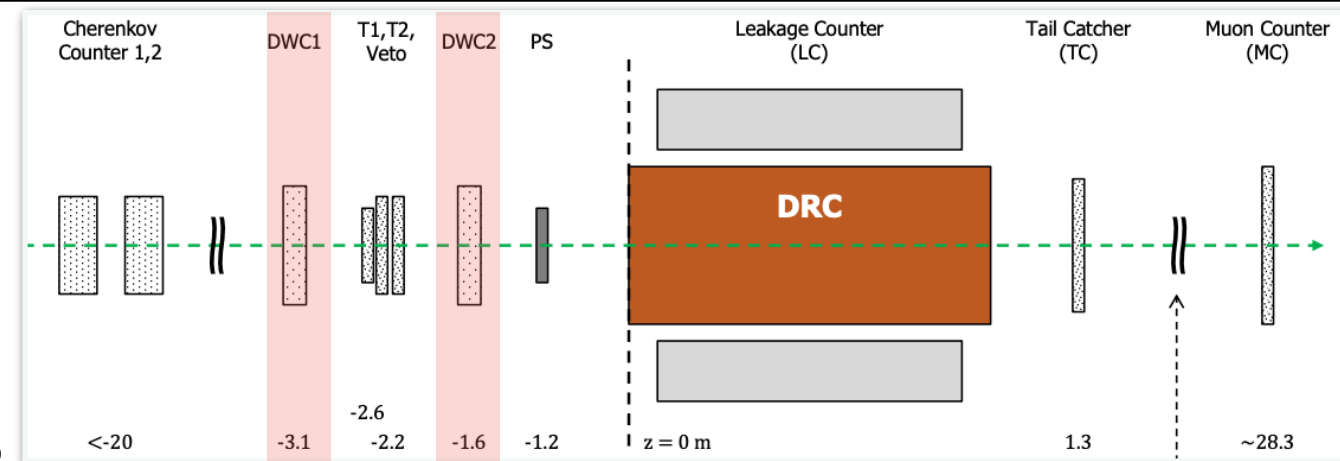
Scintillation (S) channel



Event Selection: Delayed Wire Chamber (DWC)

• DWC: Control the **beam shape** in front of the DRC modules

- Measure the 2D position (x, y) of the incident beam
- Two DWC detectors: DWC1 and DWC2
 - The **correlation** requirement between the positions of DWC1 and DWC2: powerful to suppress the **inclined** beam

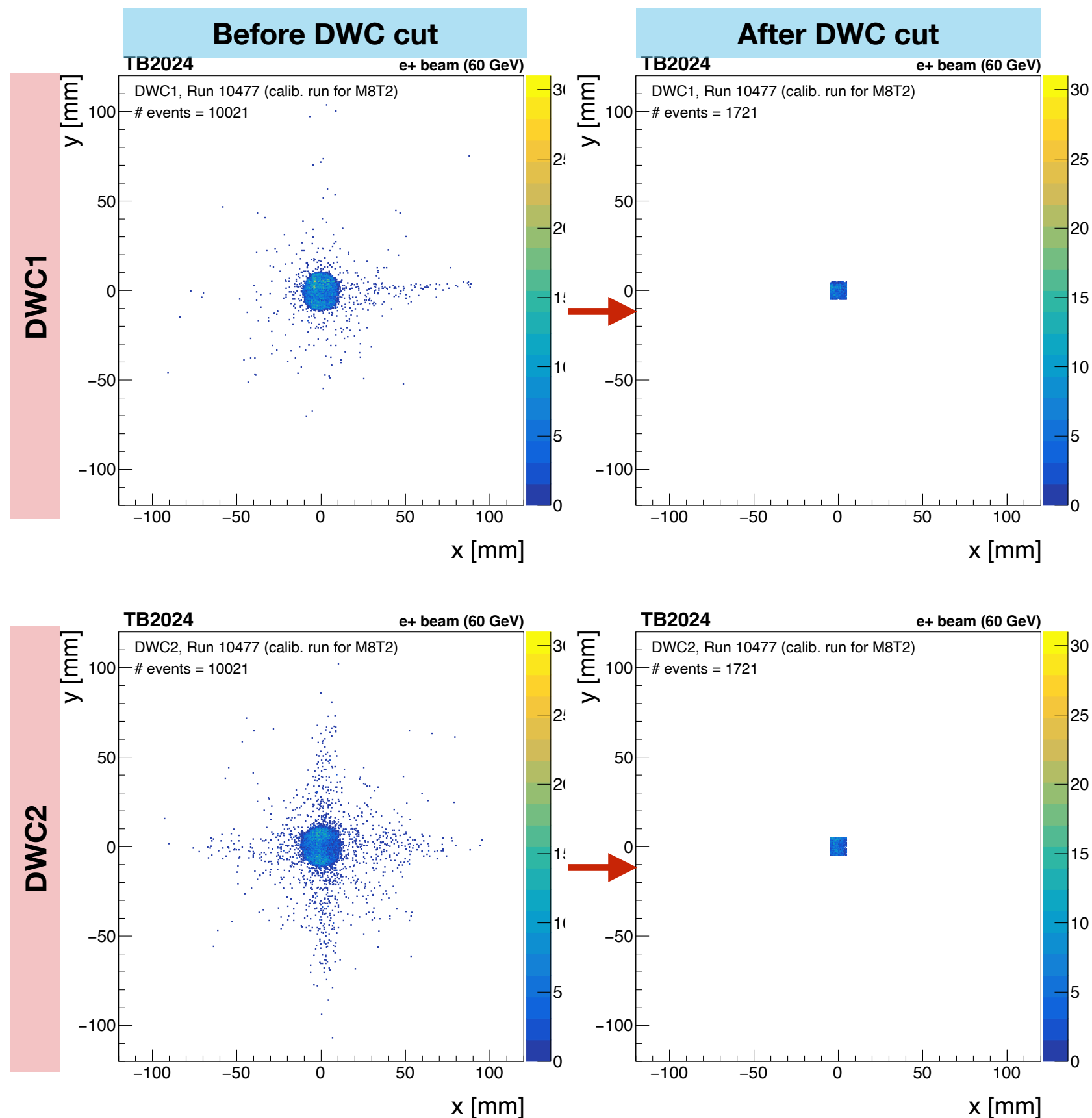


• Event selection with DWC: select the events with the beam **directed to the DRC center**

Event selection criteria for DWC	
Variable	Criteria
size, DWC1	$-5 < x < 5$, $-5 < y < 5$ mm (10 mm × 10 mm square)
size, DWC2	$-5 < x < 5$, $-5 < y < 5$ mm (10 mm × 10 mm square)
Correlation (DWC1, DWC2)	$ x^{\text{DWC1}} - x^{\text{DWC2}} < 4$ mm $ y^{\text{DWC1}} - y^{\text{DWC2}} < 4$ mm

Comparison before vs. after cut

- Example of a run with ~10k events
 - Directed to M8-T2
- The cuts are applied as intended
- ~17% events survived (10021 → 1721)

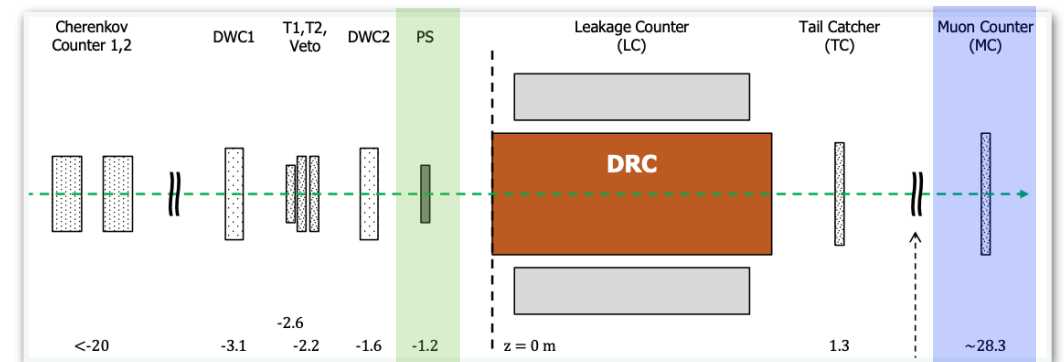


● Preshower: induce a shower for e^+ in front of the DRC

- $e^\pm$: (e^+ , e^-) pairs from the shower
→ 3, 5 ... minimum ionizing particle (mip) peak
- $\mu^\pm$ & hadrons: hardly interact → 1 mip peak
- **Select > 3 mip peak** to reject muons & hadrons

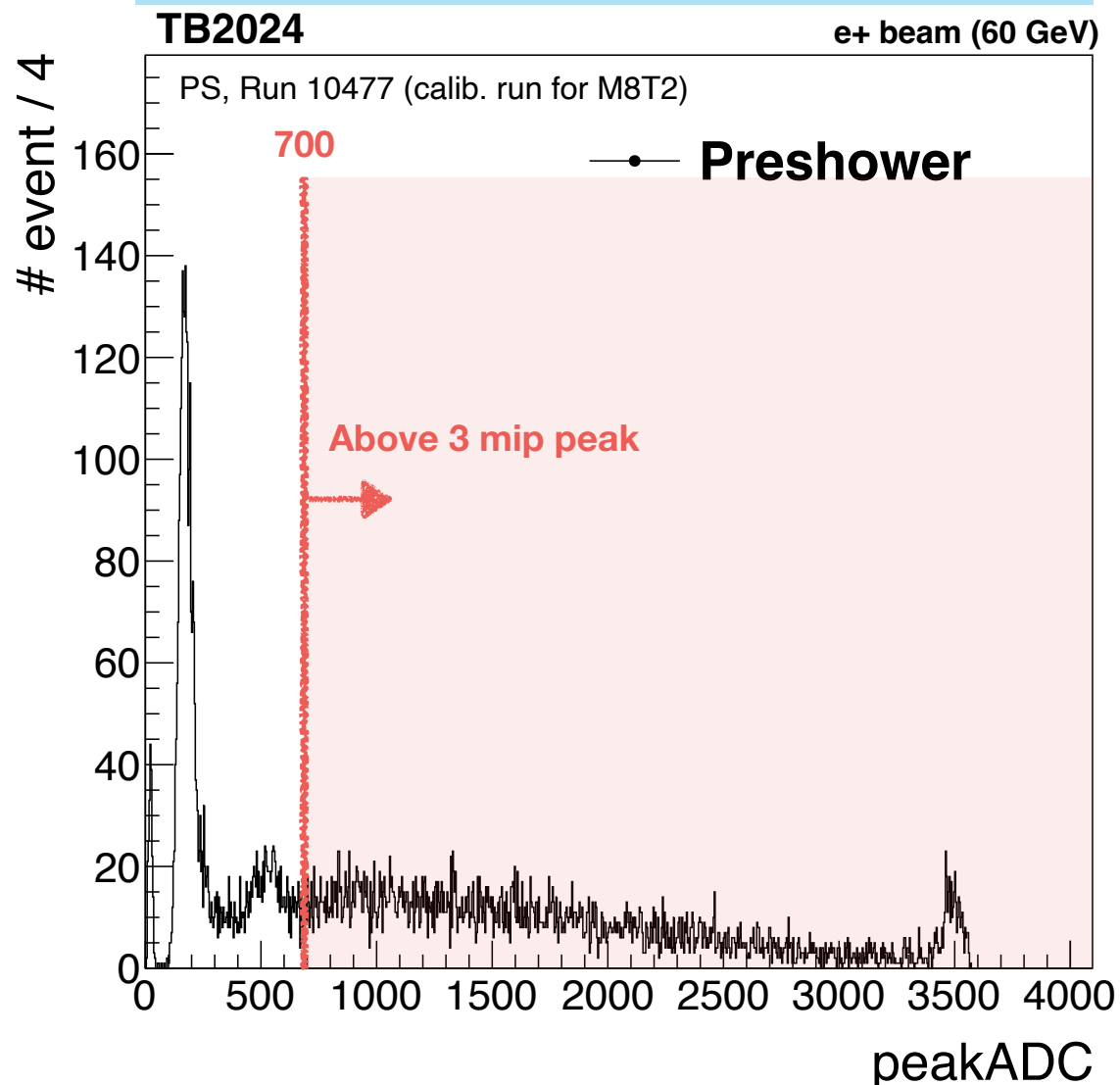
● Muon counter: make a signal if muon is detected

- **Select pedestal** of the muon counter to reject muons

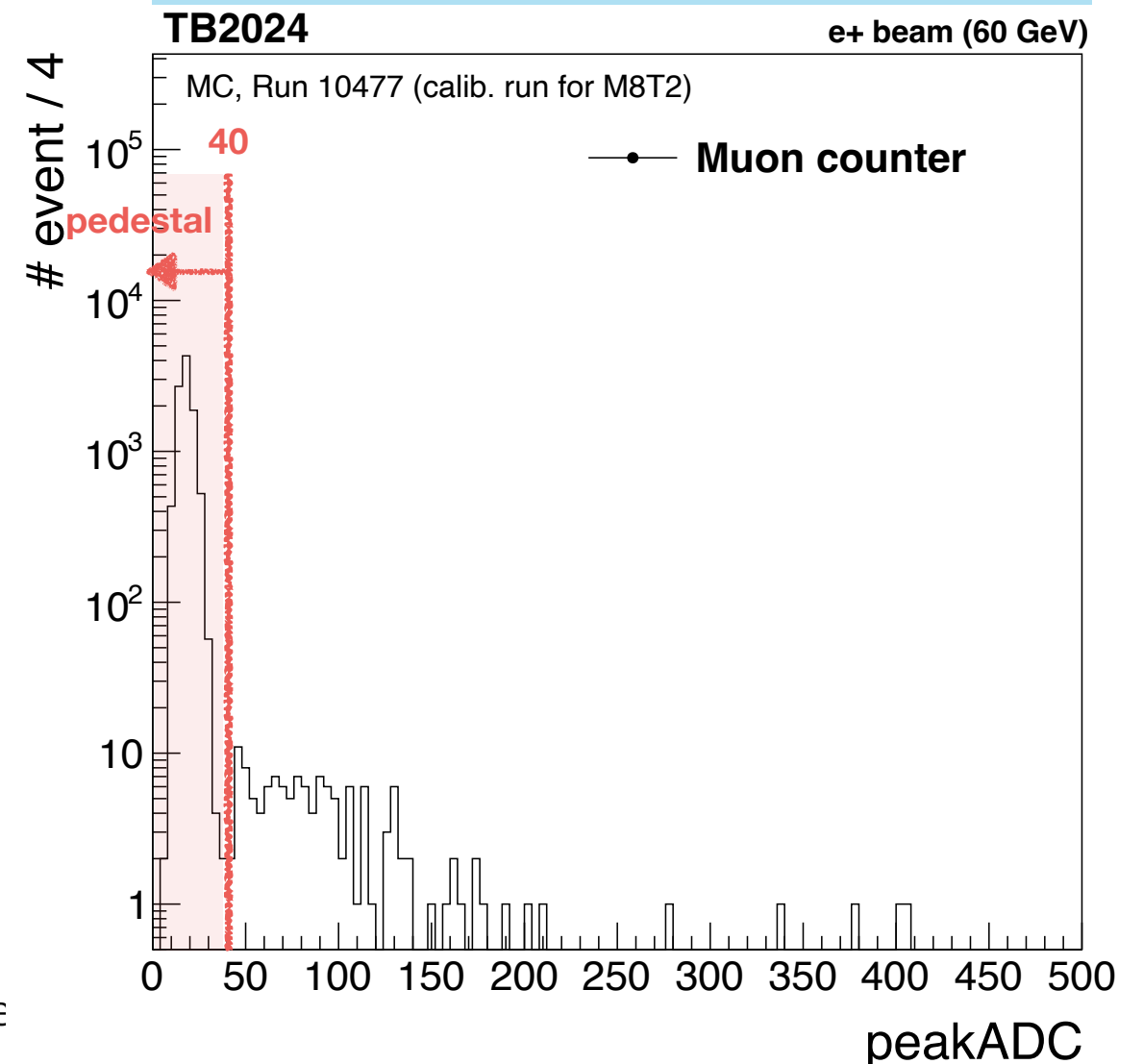


Example of the peak of ADC ("peakADC") distribution without any cut applied

Preshower



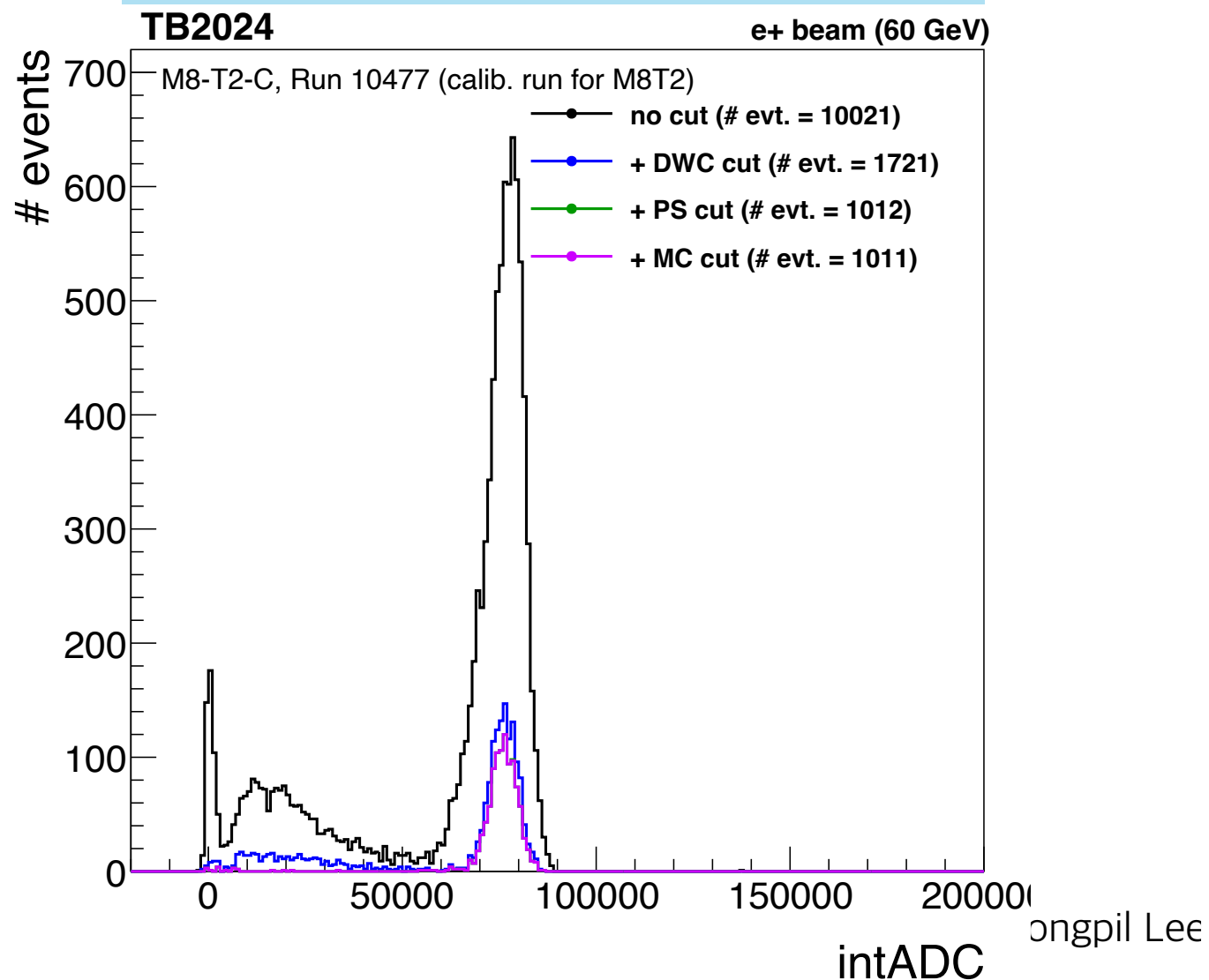
Muon counter



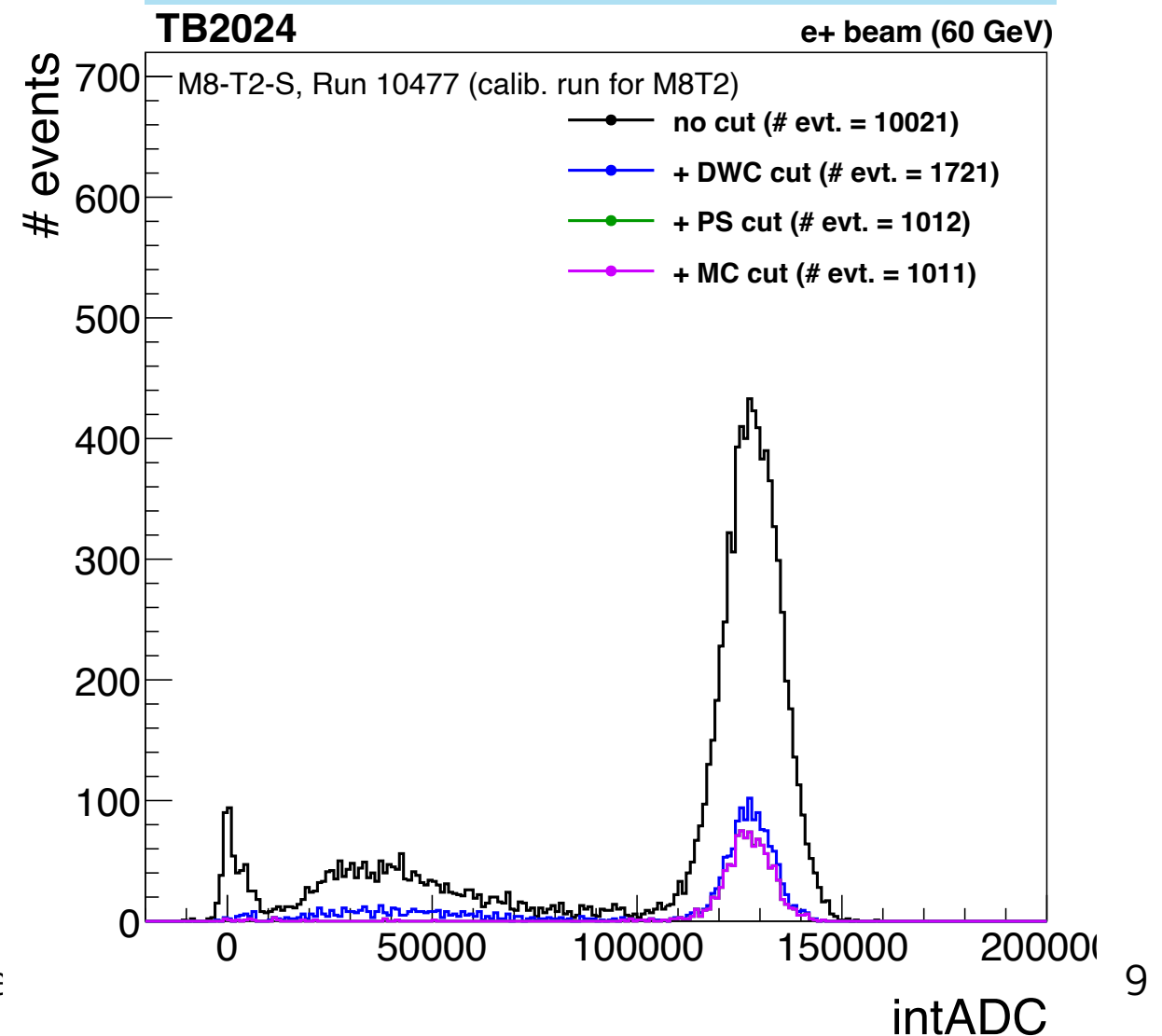
- Huge effect of **DWC** requirements (tight selection was applied)
- **Residual** backgrounds (muons & hadrons) below the peak region: **rejected by PS & MC**
- **~10%** events are survived after the event selection

Example of IntADC distribution @ Module 8, Tower 2 (M8-T2): ~10k events

C channel

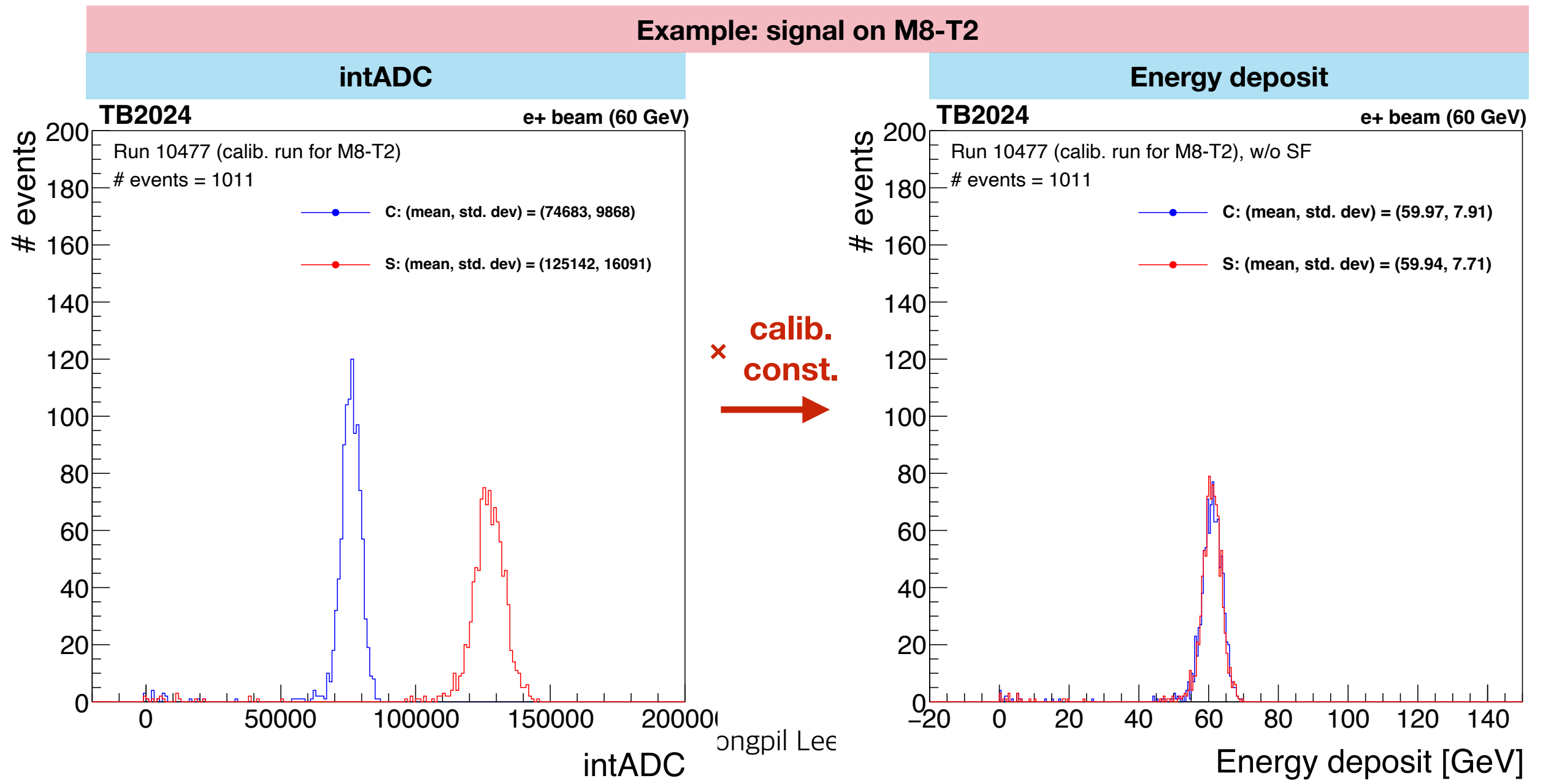


S channel



Calibration: Convert from IntADC to Energy Deposit

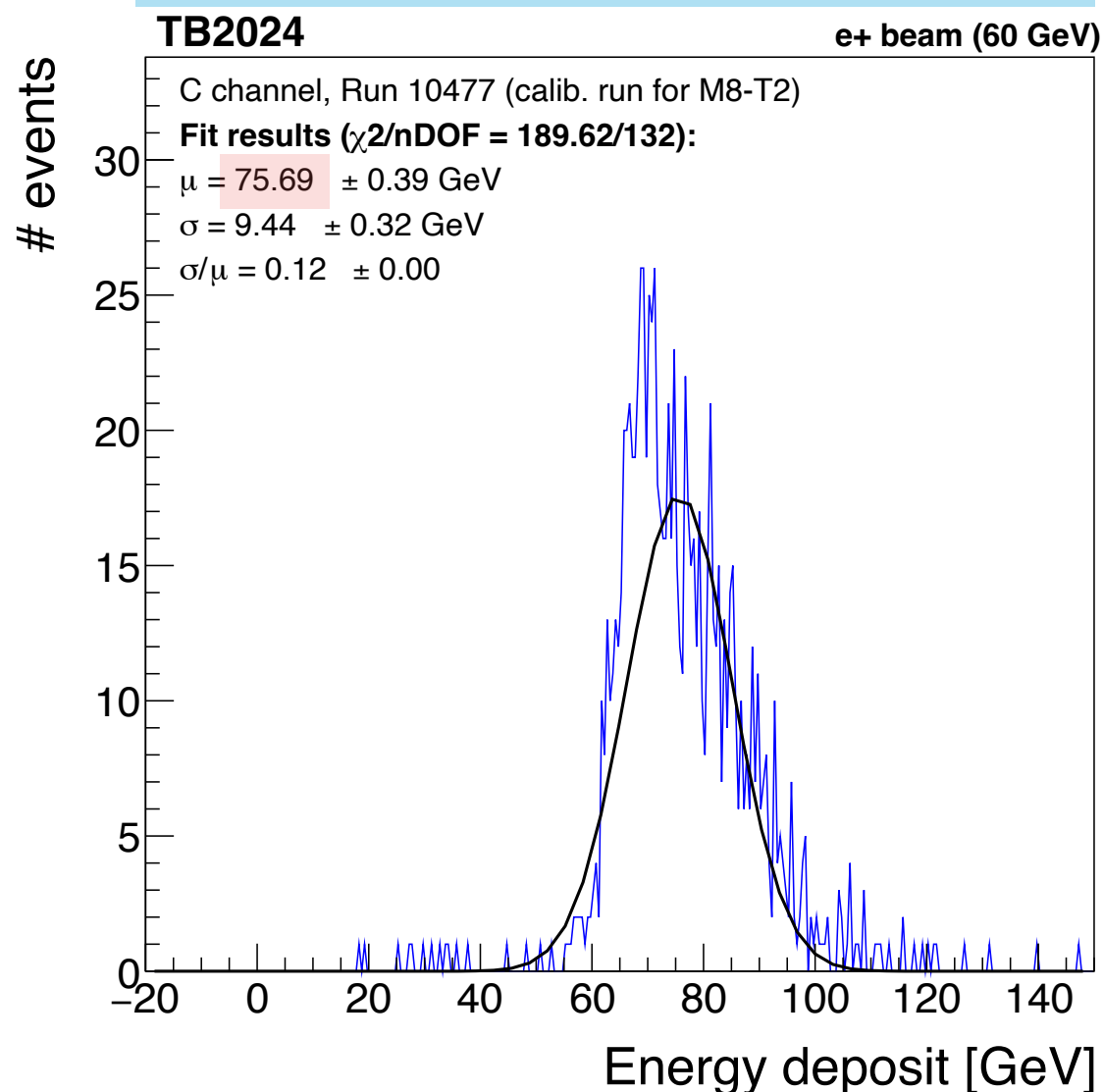
- Using the data with 60 GeV e⁺ beam for each tower (total 36 runs)
 - Observed intADC distribution: corresponds to 60 GeV energy deposit
→ Calibration constant (conversion factor) = **60 GeV / mean(intADC) for each tower**
- Before vs. after applying the calibration constant: M8-T2 as an example
 - Mean value is close to 60 GeV as expected



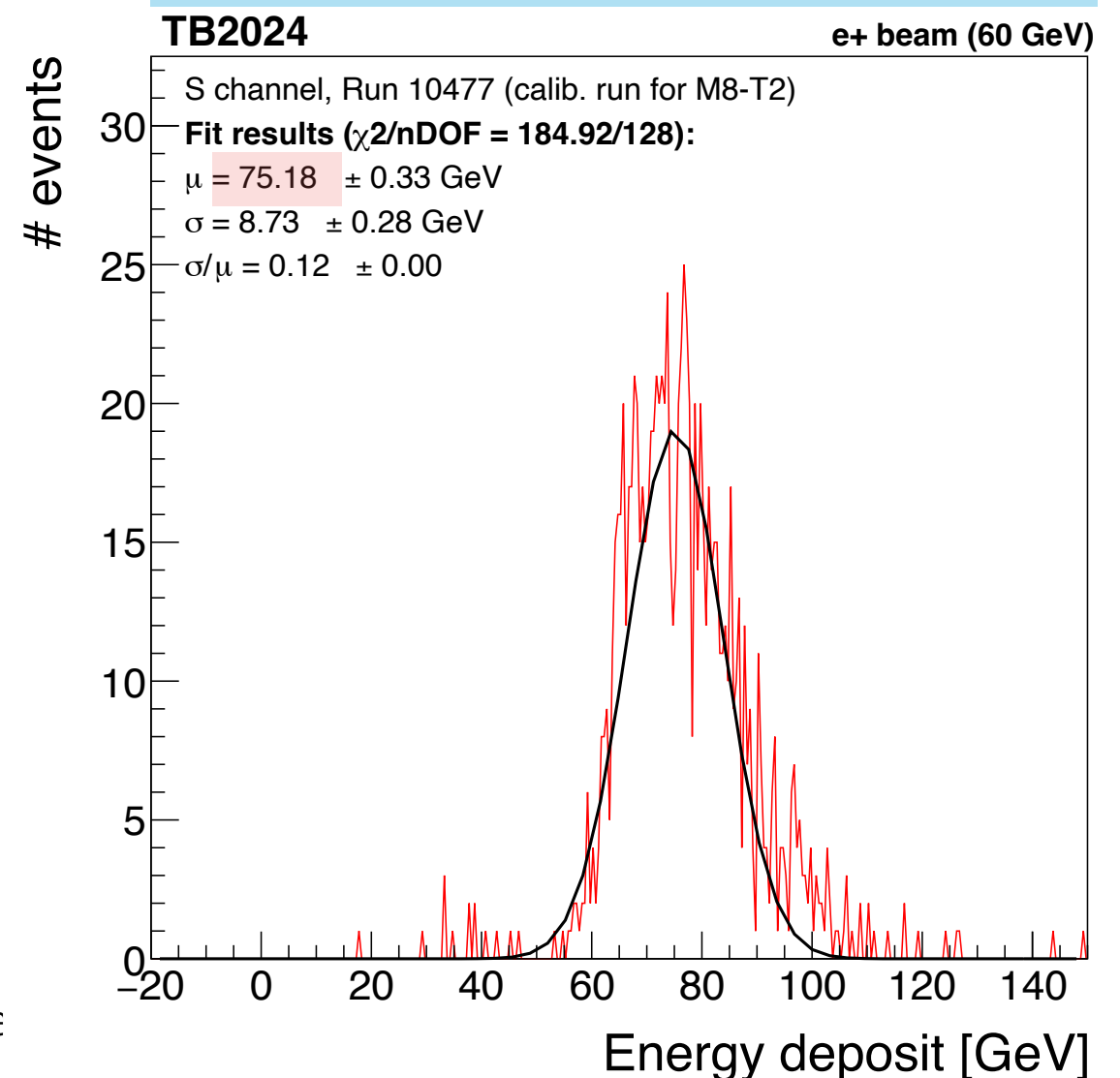
- Match the total energy deposit on DRC to the expected deposit on the DRC size
- Scale factor = **59.52 GeV / mean(total energy deposit in the DRC)**
 - $59.52 \text{ GeV} = 60 \text{ GeV} * 99.2\%$
 - 99.2%: expected energy deposit fraction from the simulation
 - $SF(C) = 59.52 / 75.69 \sim \mathbf{0.786}$; $SF(S) = 59.52 / 75.18 \sim \mathbf{0.792}$

Total energy deposit to the DRC (energy sum of all towers) before SF

C channel



S channel



Energy Scan Results (Center: M8-T2)

※ Other energy points: backup

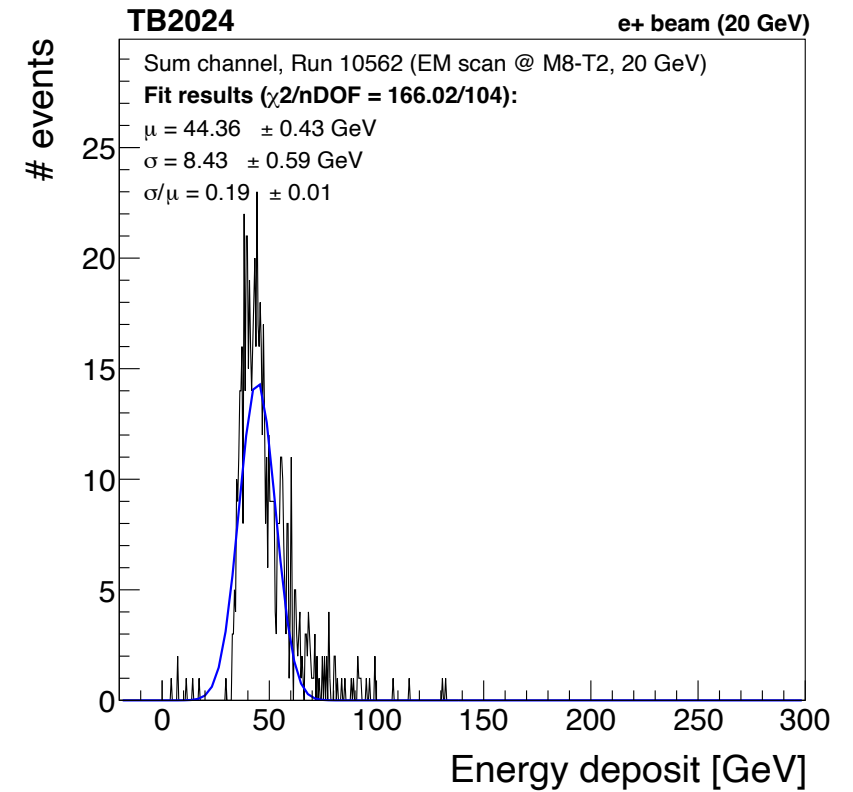
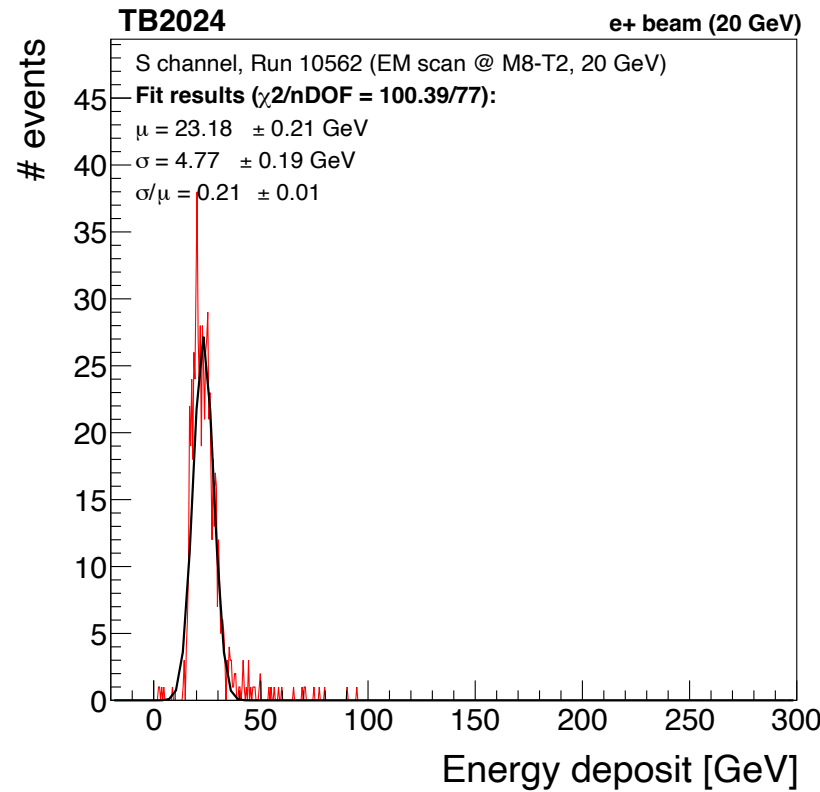
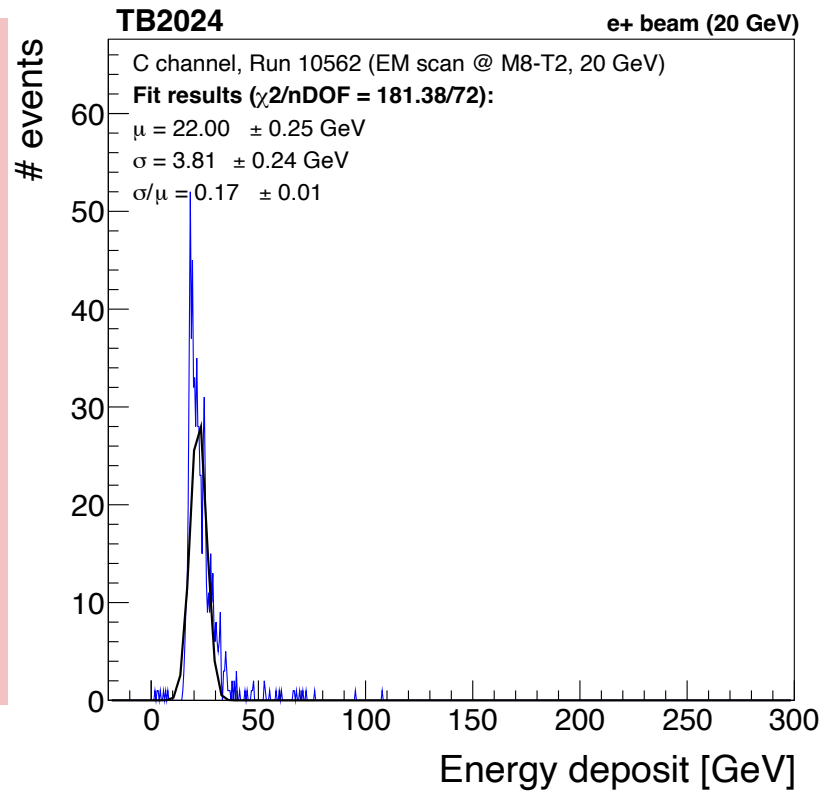
✓ Two representative cases (20 and 80 GeV): Follow the **gaussian** distribution as expected

C channel

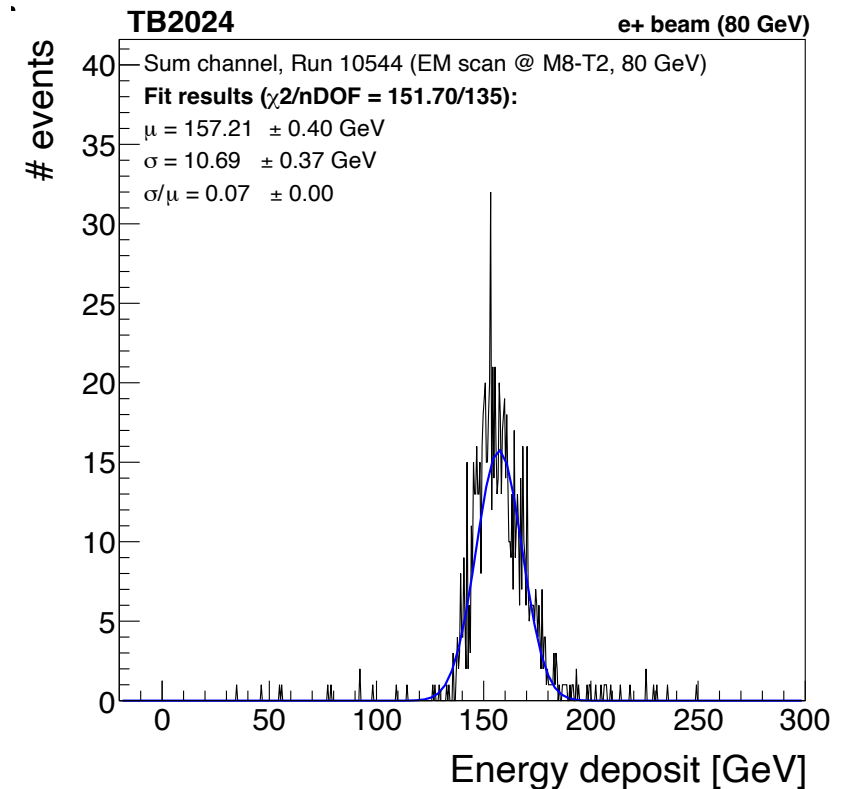
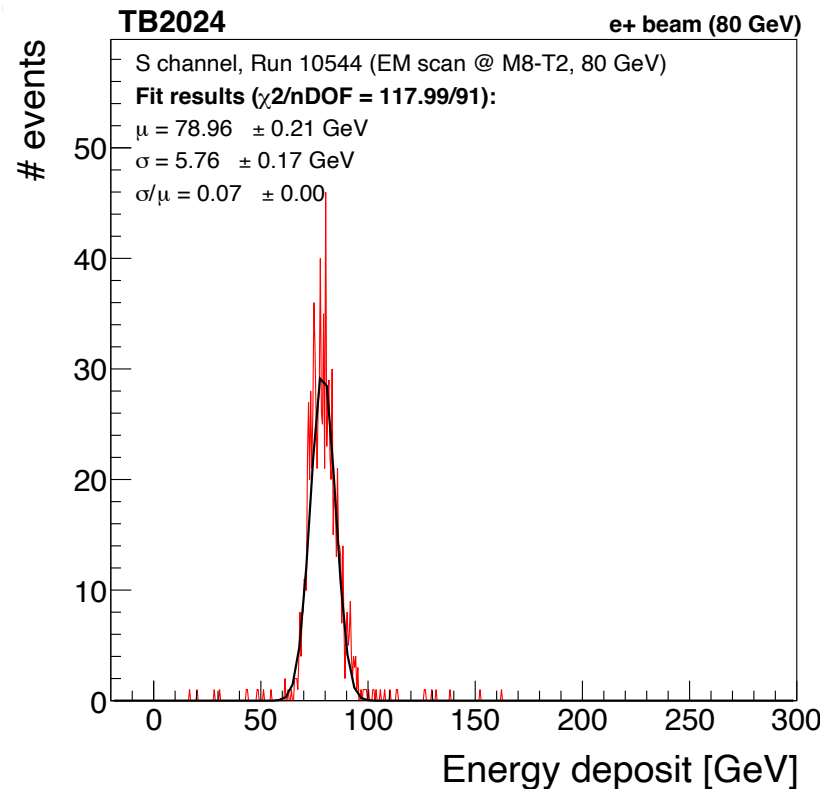
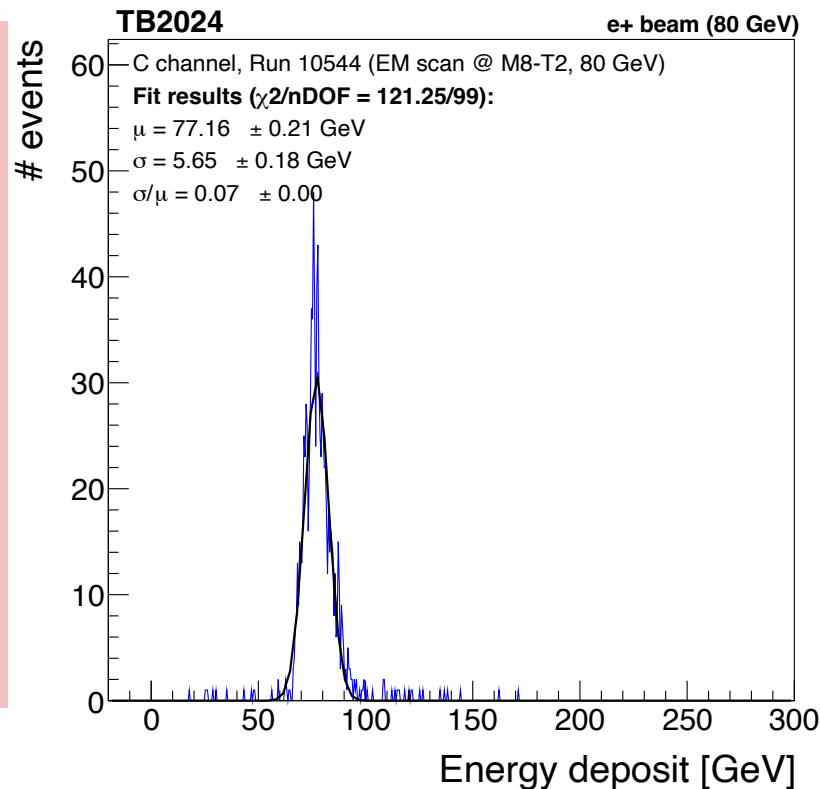
S channel

Sum (C+S)

20 GeV



80 GeV



Linearity

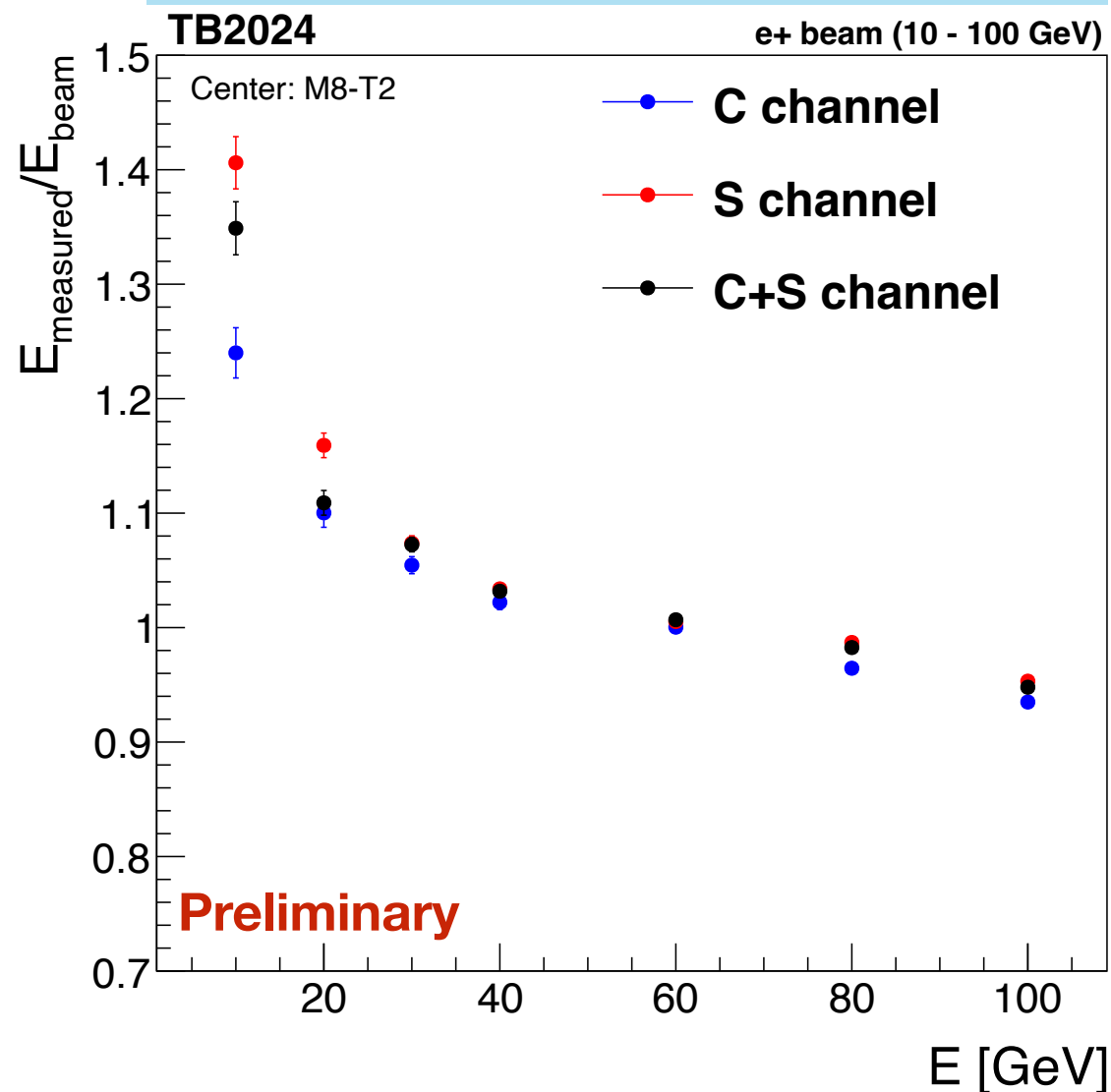
- Overestimated in the low energy; closer to 1.0 in the higher energy

EM resolution vs. energy

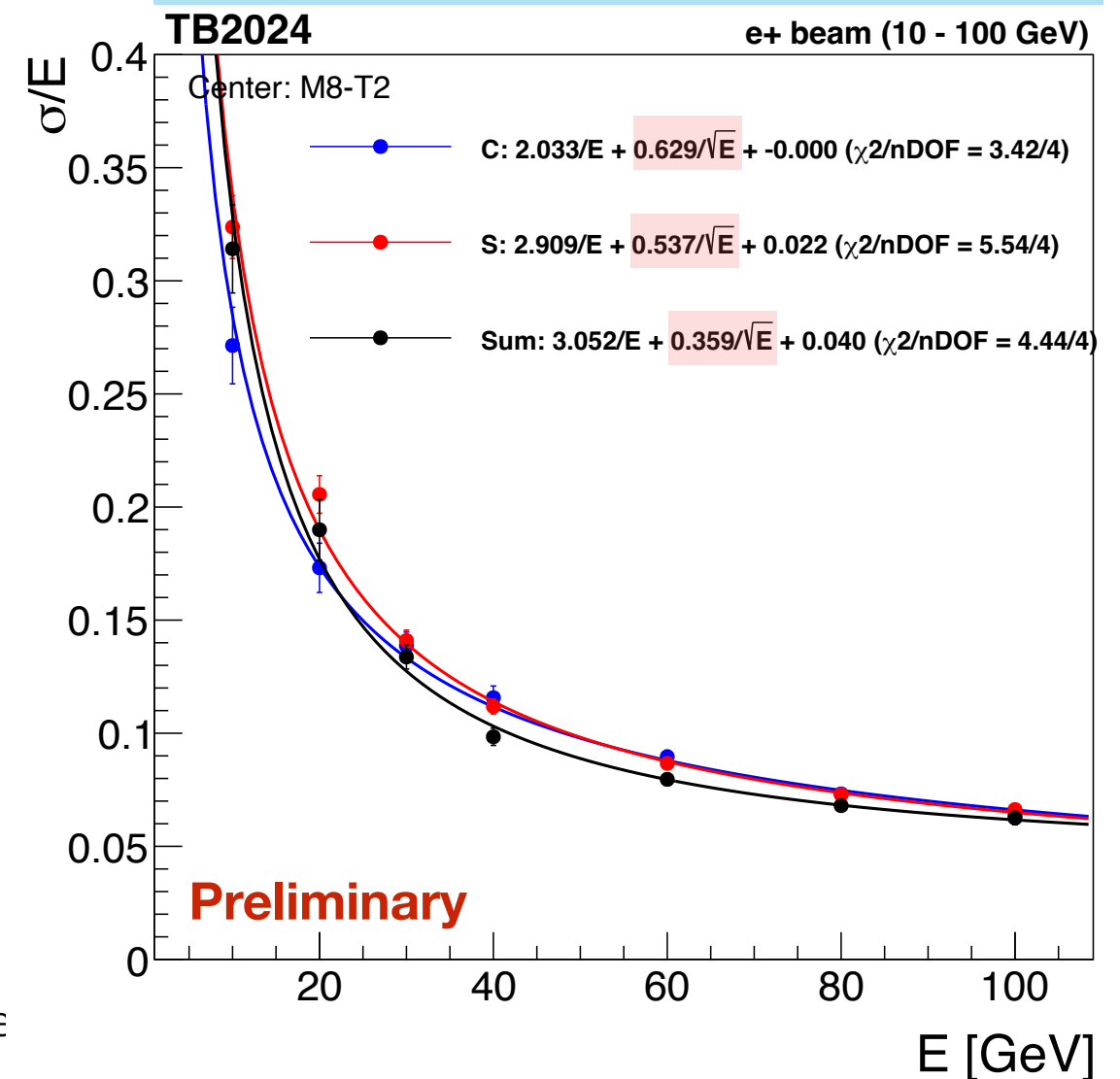
- Fit with the expected scaling in $1/\sqrt{E}$, with the error term $1/E$ and constant term
- Better resolution in the sum of C and S channels as expected
- Worse performance than expected in general (expected ~20% (or below) stochastic terms)

EM resolution & linearity results

Linearity



EM resolution vs. E



- Reproduce the results using the **9 towers near the center only**

- To check the effect from the noise from the towers with small signals

- Scale factor is re-calculated with respect to the 9 towers (97.3% deposit)

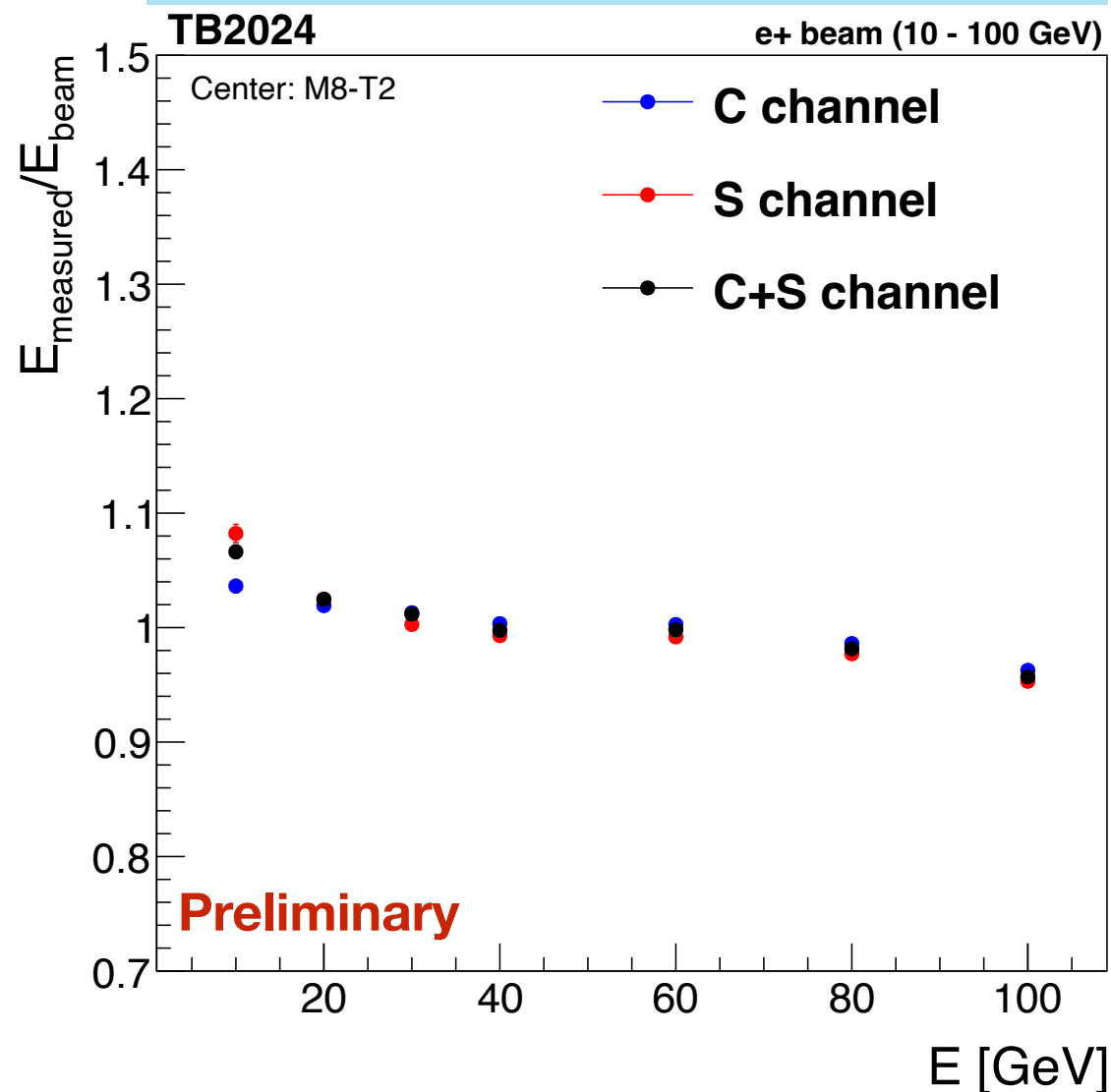
- Much better linearity & EM resolution results

- Indicate that the noise from the towers with the small signals is significantly worsening the results

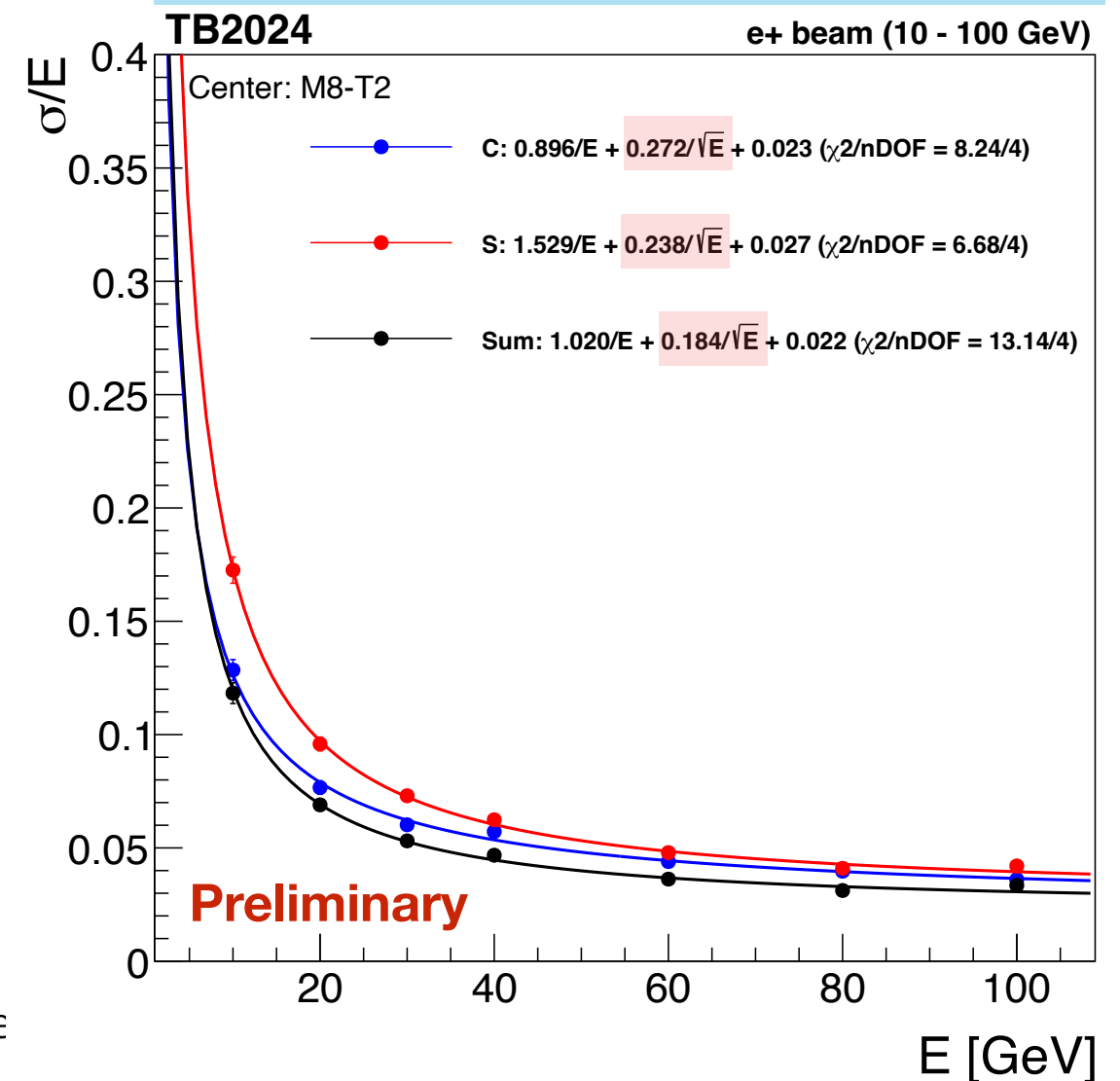
M1	M2		M3		
M4	T1	T2	T1	T2	M6
	T3	T4	T3	T4	
M7	T1	T2	T1	T2	
	T3	T4	T3	T4	
	M8		M9		

EM resolution & linearity results

Linearity



EM resolution vs. E



Summary

- ◉ The **electromagnetic** performance of the dual readout calorimeter is presented
 - Using the data collected in the test beam experiment at CERN SPS in 2024
 - **Integration range** was determined to cover the ADC signals from DRC fibers
 - **Event selection** was performed to keep the pure e^+ beam events
 - Using DWC, PS and MC auxiliary detectors
 - **Calibration** was performed to convert intADC to energy deposit for each tower
 - Preliminary **linearity & EM resolution vs. energy** were presented
 - Overestimate the beam energy in the low energy; agree within 2-4% in the higher energy region
 - EM resolution is following the expected shape in general; but with worse performance than expectation
 - The performance was largely improved with the towers near the center only
 - Indicate **the noise effect** from the far-towers with small signal (pedestal fluctuation)
- ◉ Next step
 - Investigation on the noise effect & find a way to improve the results if possible
 - In terms of the time structure & pedestal distributions, etc
 - Develop a strategy to reduce the noise effect for the next test beam experiments



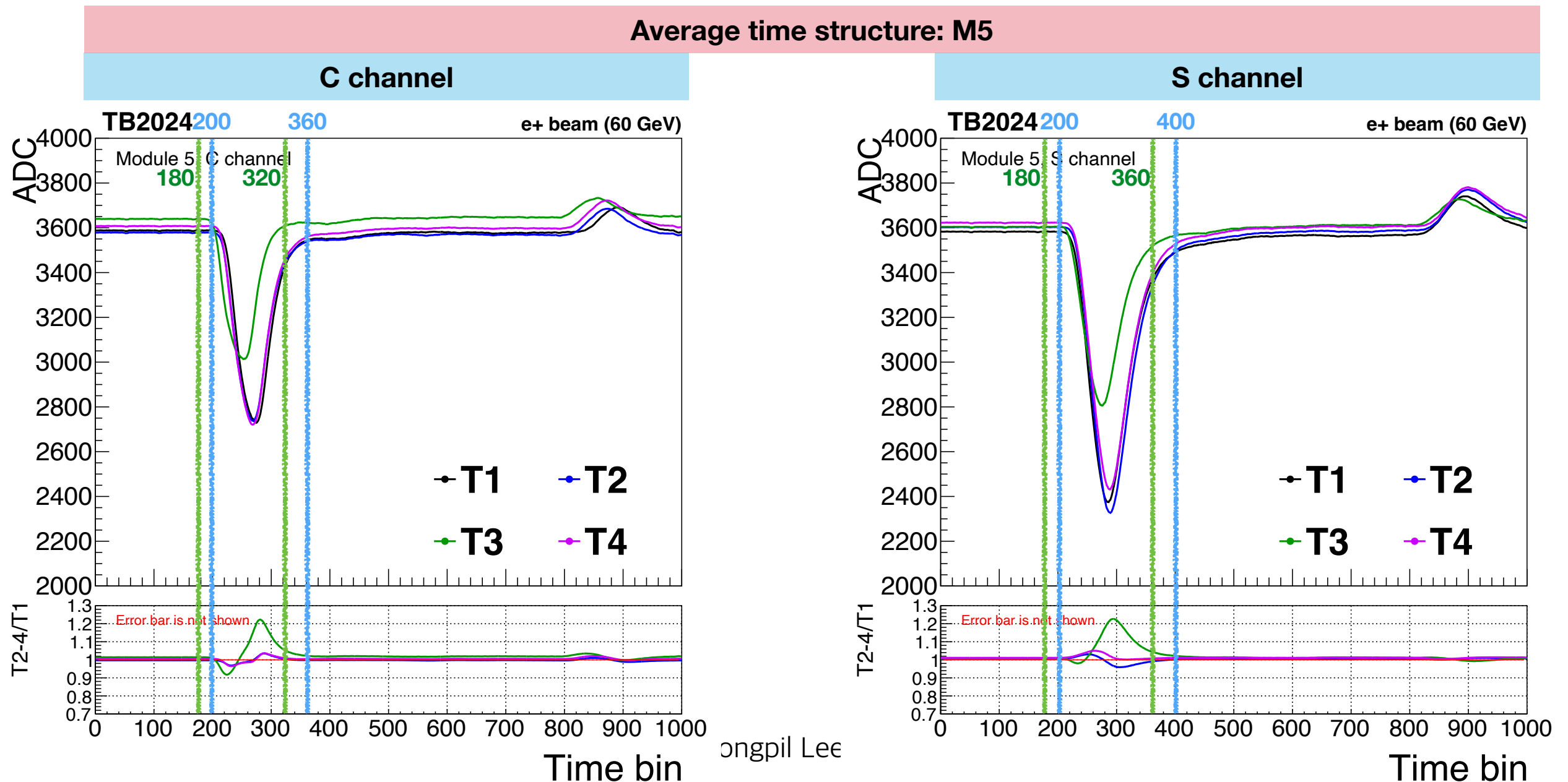
Backup

- Reference: [Nucl. Instrum. Methods Phys. Res., Sect. 735, 130 \(2014\)](#), [Rev. Mod. Phys. 90, 025002 \(2018\)](#)

- Resolution formula: $\frac{\sigma}{E} = \frac{a_{\text{stoch}}}{\sqrt{E}}$ with $a_{\text{stoch}} = a_{\text{samp}} \oplus a_{p.e.}$

- Stochastic** fluctuation is the dominant source for the resolution of the calorimeter
 - Consist of two sources: **sampling** fluctuation (a_{samp}) and **photoelectron** fluctuation ($a_{p.e.}$)
- [1] **sampling** fluctuation, a_{samp}
 - Origin
 - ✓ 1) Fluctuations in # of different shower particles that contribute to the calorimeter signals
 - ✓ 2) Fluctuations in the amount of energy deposited by individual shower particles in the active calorimeter layers
 - (Empirical) expression: $a_{\text{samp}} = 0.027 \sqrt{d/f_{\text{samp}}}$, d : thickness of active material (or diameter if it is a fiber)
 f_{samp} : sample fraction (ratio of active vs. passive materials)
- [2] **photoelectron** fluctuation
 - Fluctuation in the # of photoelectrons emitted by the C or S fibers
 - $a_{p.e.} = 1/\sqrt{N_{GeV}}$ where N_{GeV} = light yield (# photoelectrons emitted per GeV)

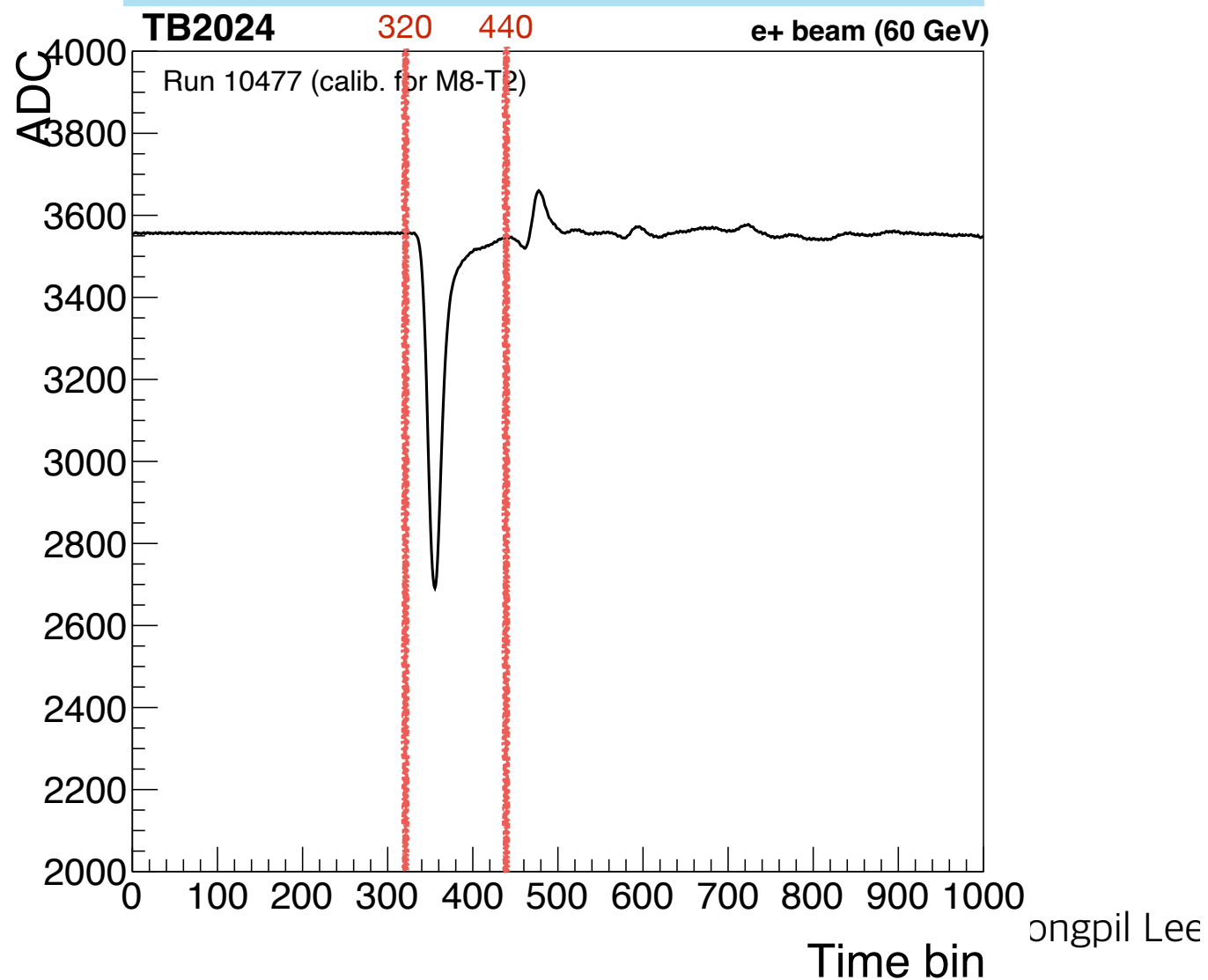
- Integration range for M5 is adjusted because of its different timing than the other modules
 - T1, T2, T4: [200, 360] for C channel; [200, 400] for S channel
 - T3: [180, 320] for C channel; [180, 360] for S channel
 - Different timing characteristic due to its special PMT (MCP-PMT)



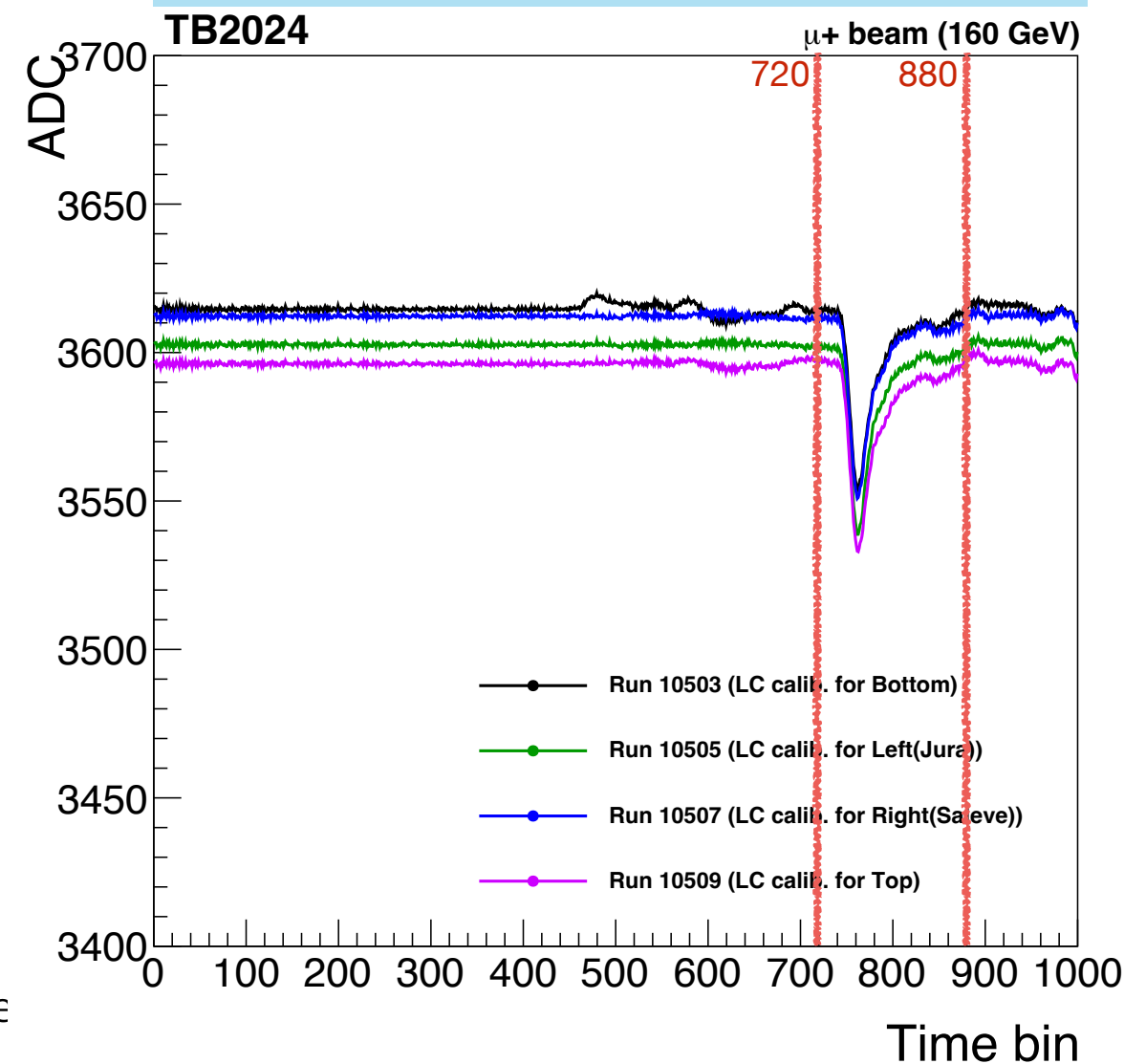
- Determine the range for aux. detectors by checking the structure as well
 - Remark: only peakADC will be used → not sensitive to the fine tuning of the range
 - **Preshower: [320, 440]**
 - Consistency between runs is checked (over calibration runs: backup) → **common range over all runs**
 - **Muon counter: [720, 880]**
 - Consistent over the μ^+ beam runs → **common range over all runs**

Average time structure: Auxiliary Detectors

Preshower (M8T2 Calib. run)



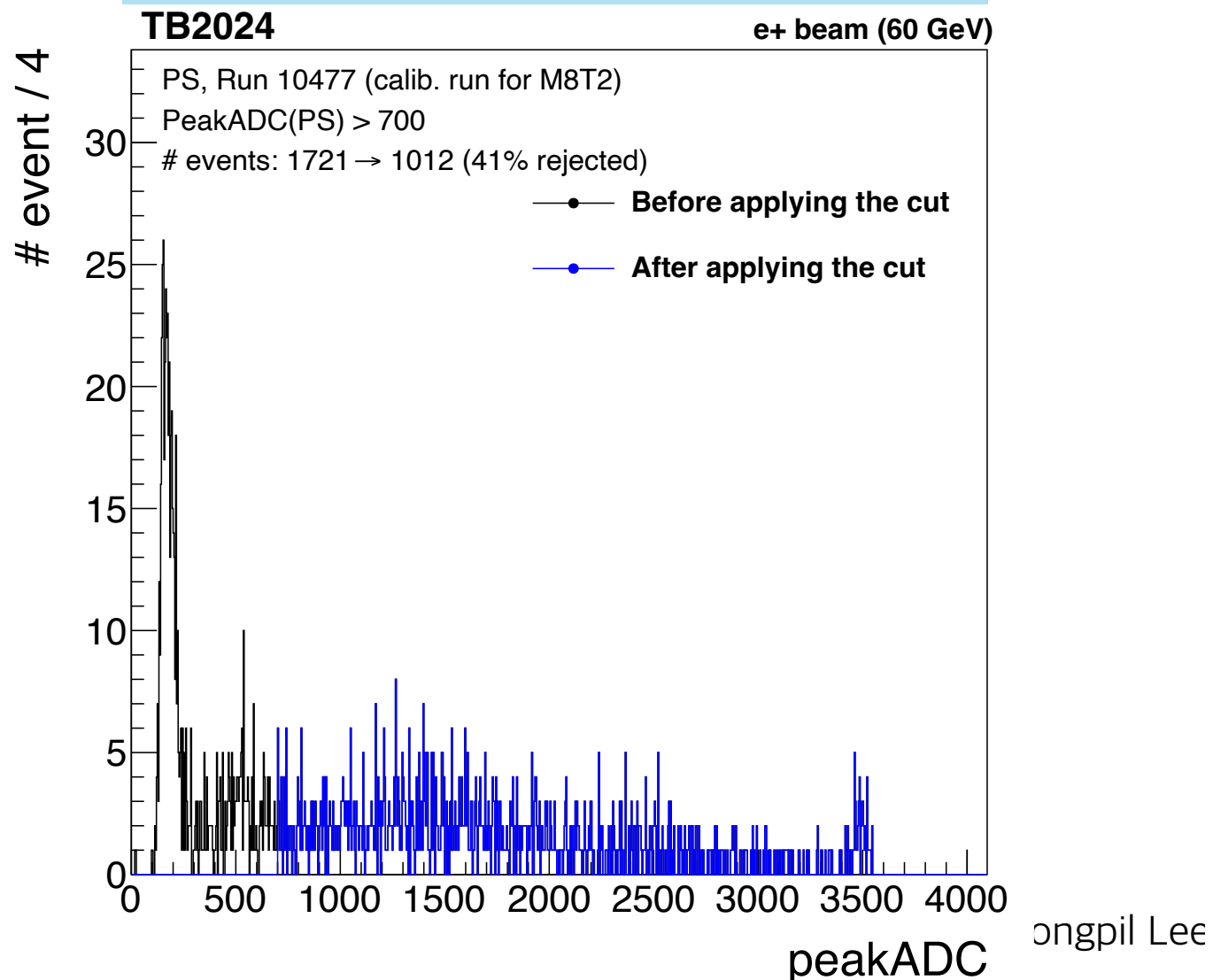
Muon counter (LC calib. run with μ^+ beam)



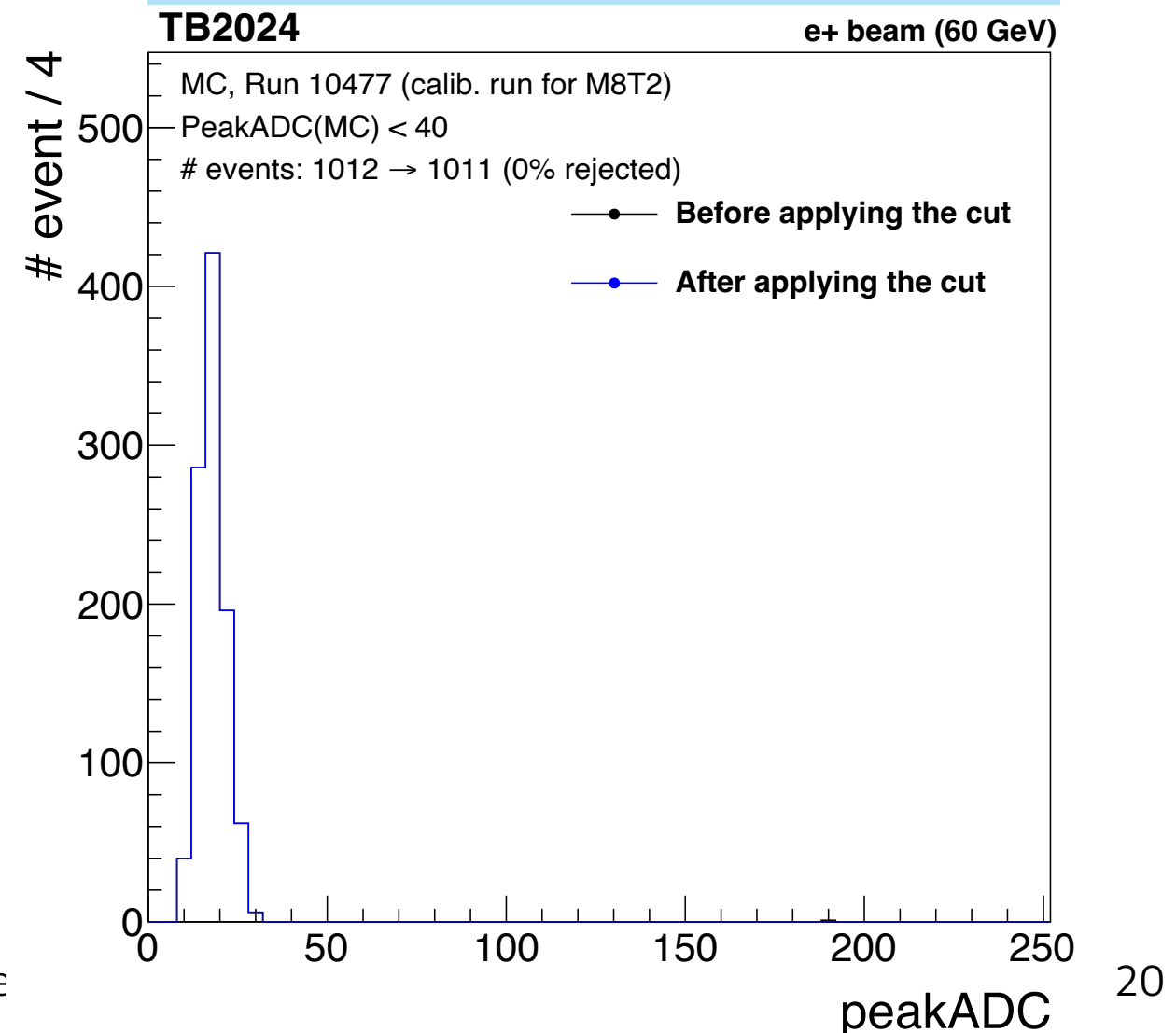
- Check the peakADC distribution before vs. after the cut applied
 - After preshower cut: ~40% rejected
 - After muon counter cut: almost no effect (already pure enough)
- Survival fraction of total event selection: ~10% (10021 → 1011 for Run 10477)

PeakADC distribution (calibration run for M8T2)

Preshower: DWC → DWC+PS

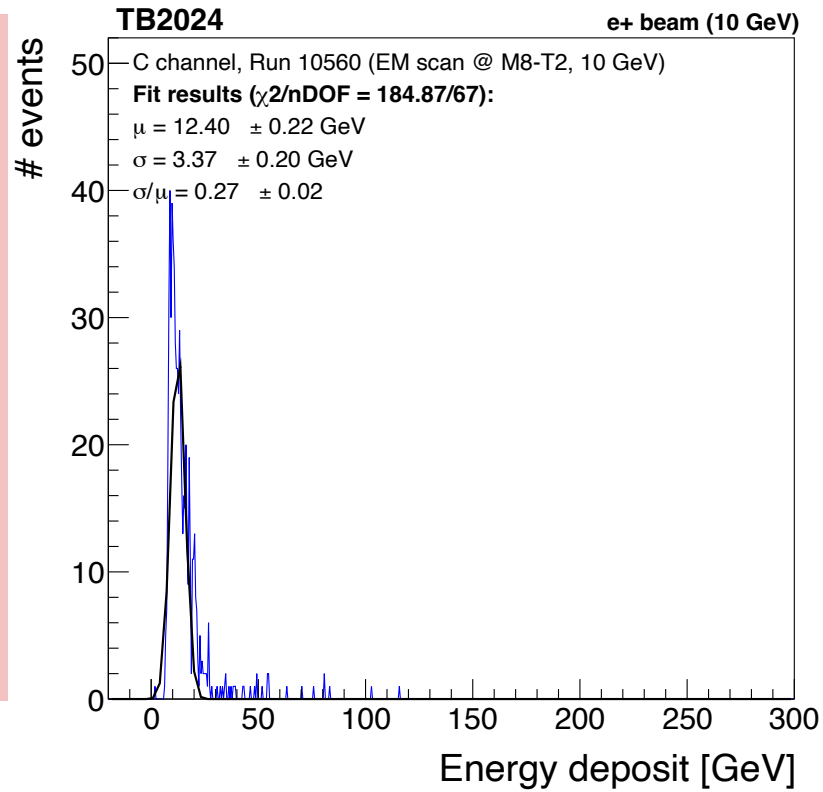


Muon counter: DWC+PS → DWC+PS+MC

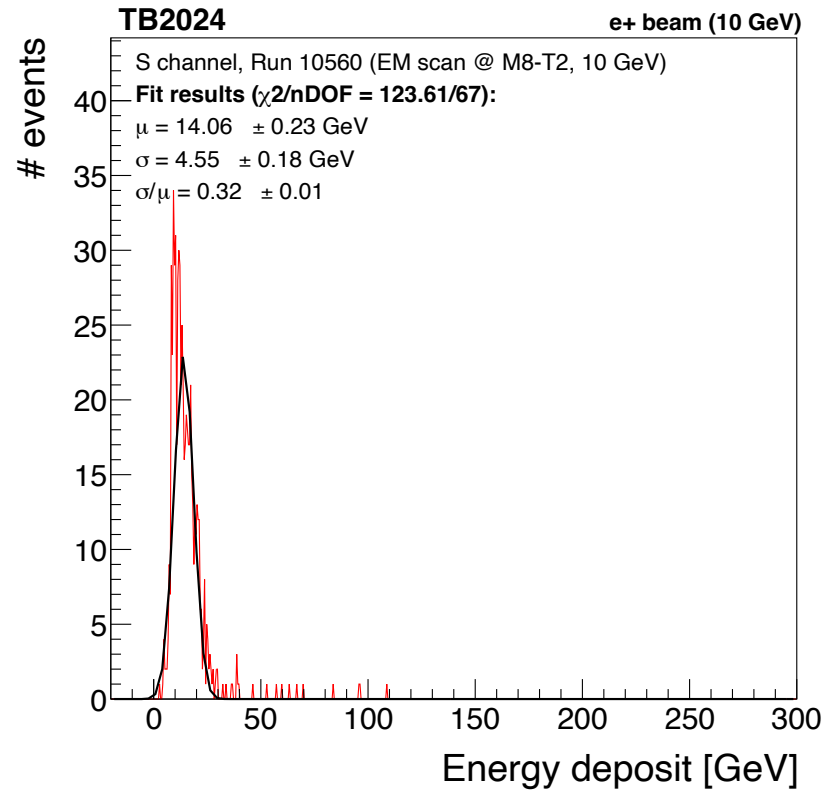


Energy Scan Results (Center: M8-T2)

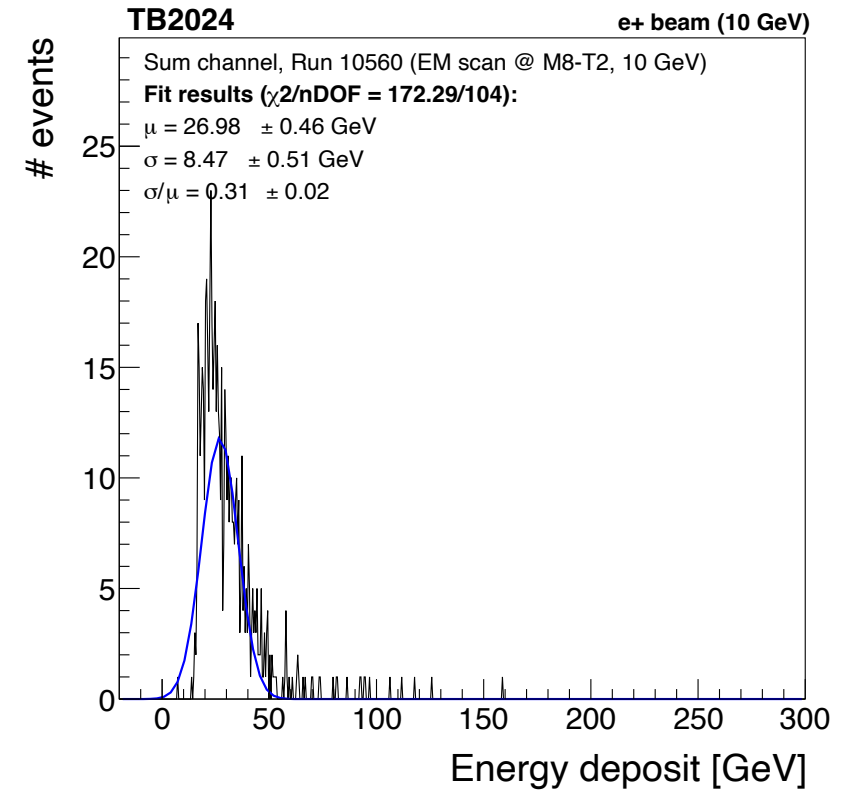
C channel



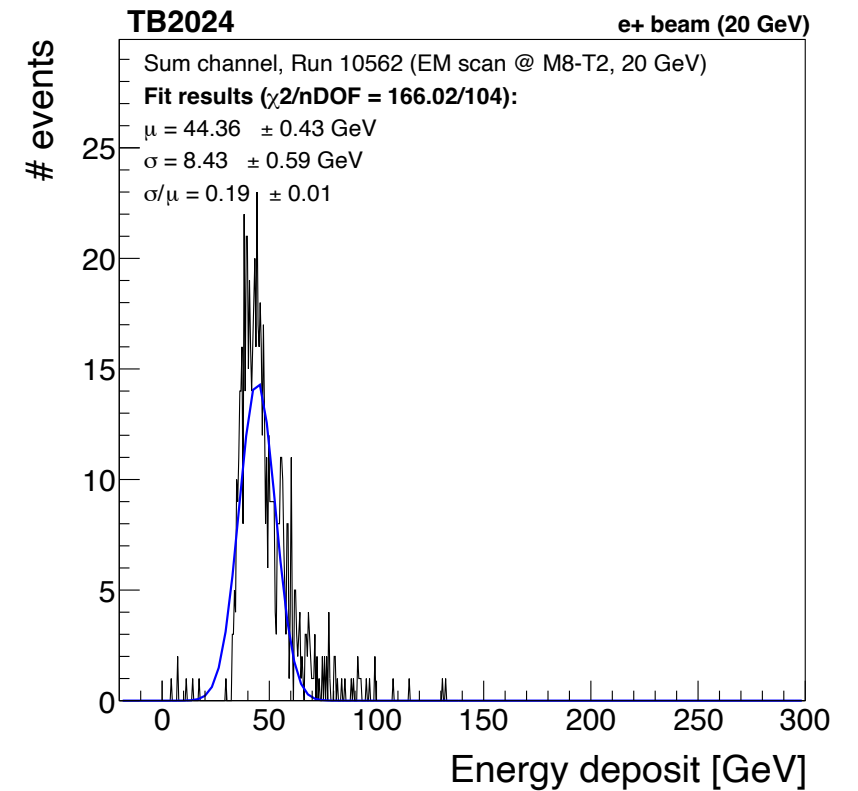
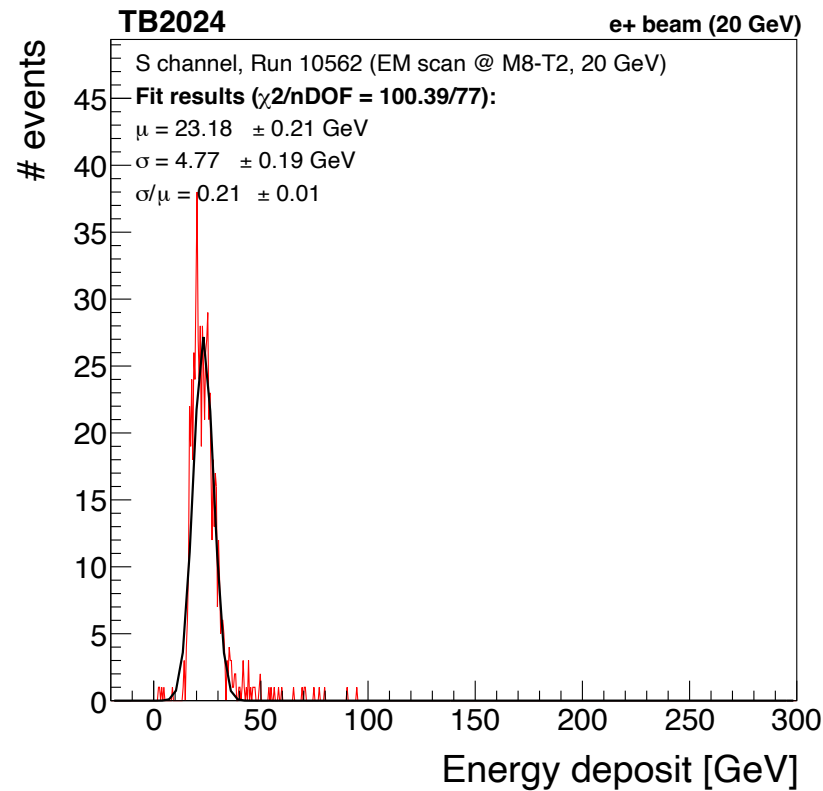
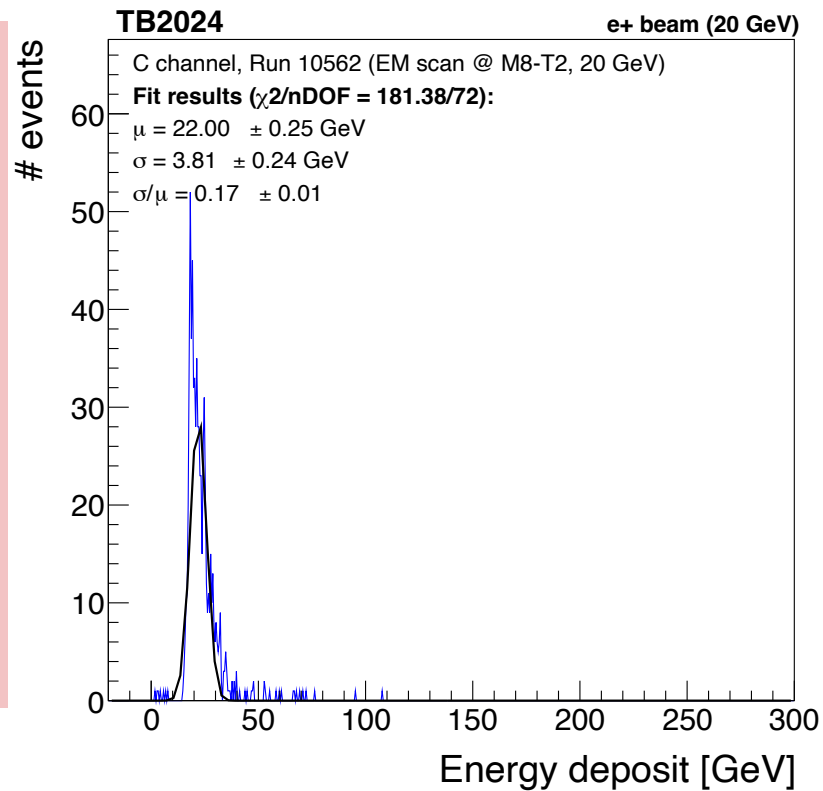
S channel



Sum (C+S)

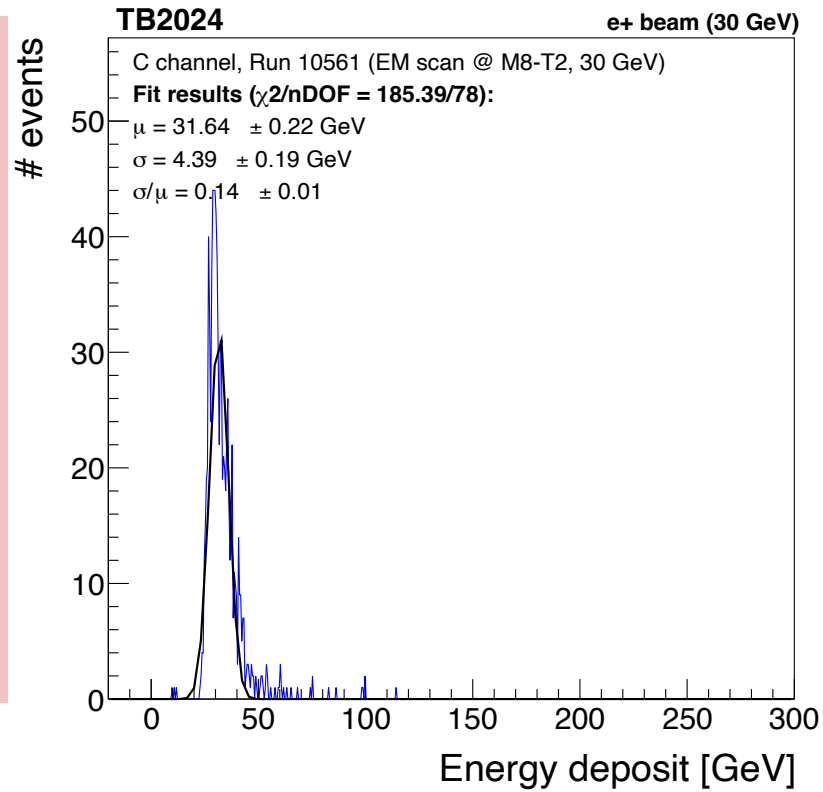


20 GeV

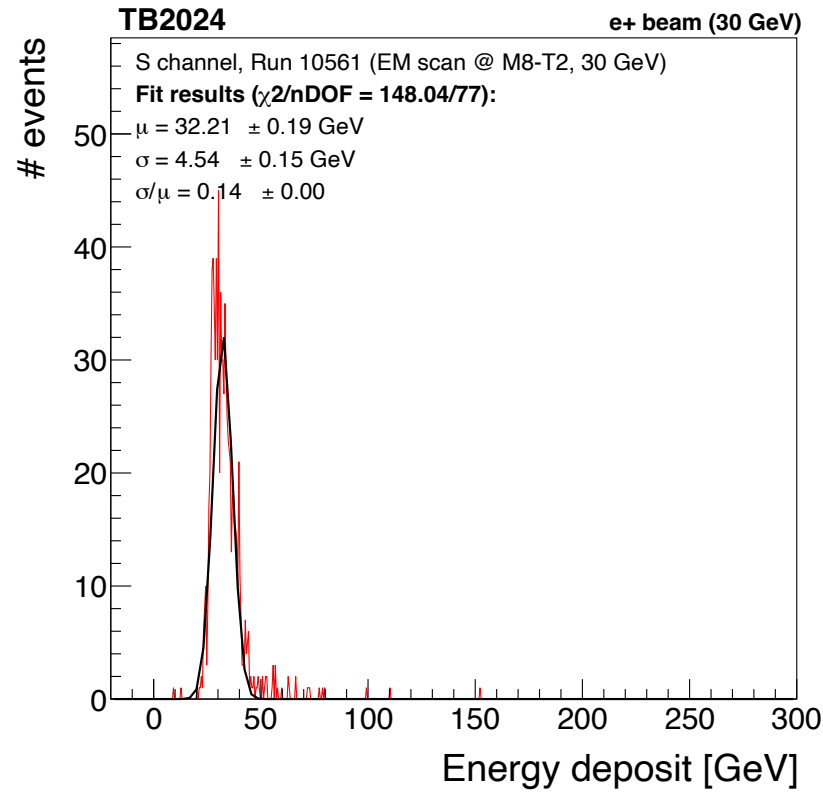


Energy Scan Results (Center: M8-T2)

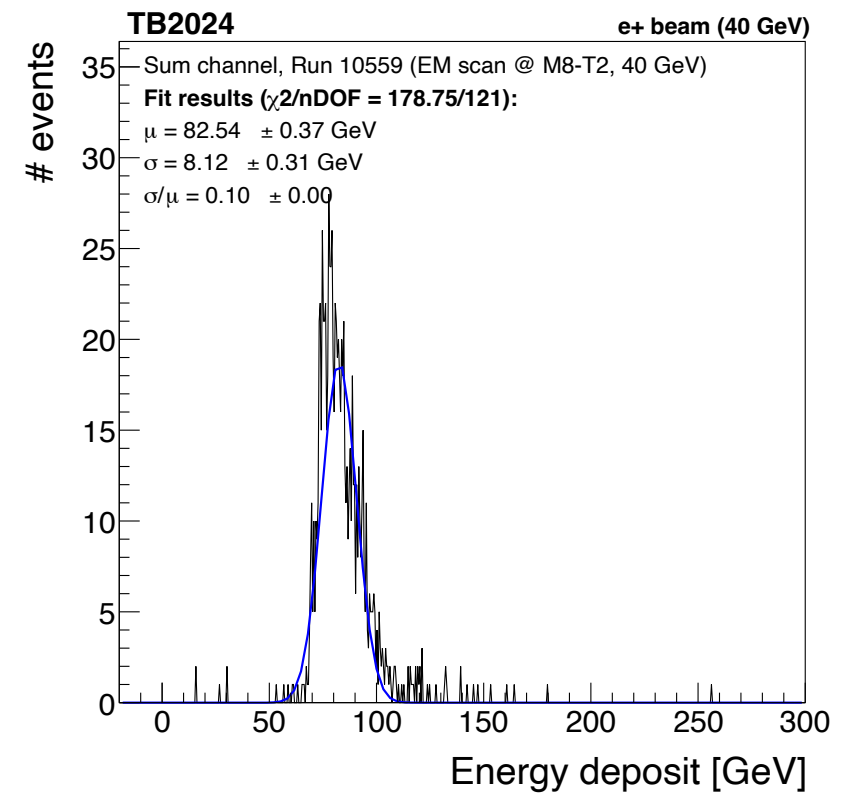
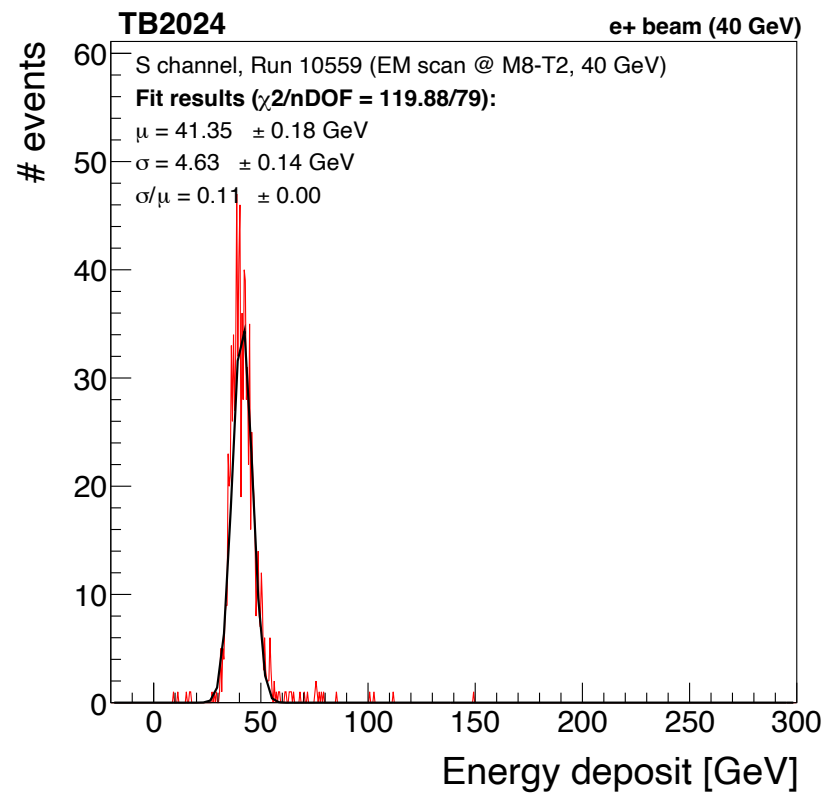
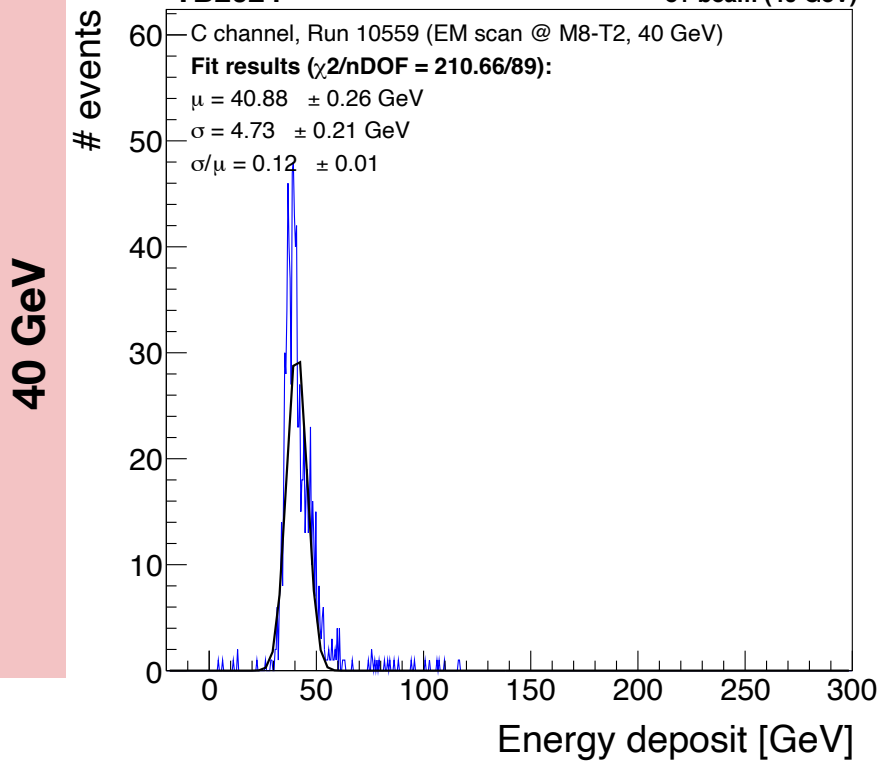
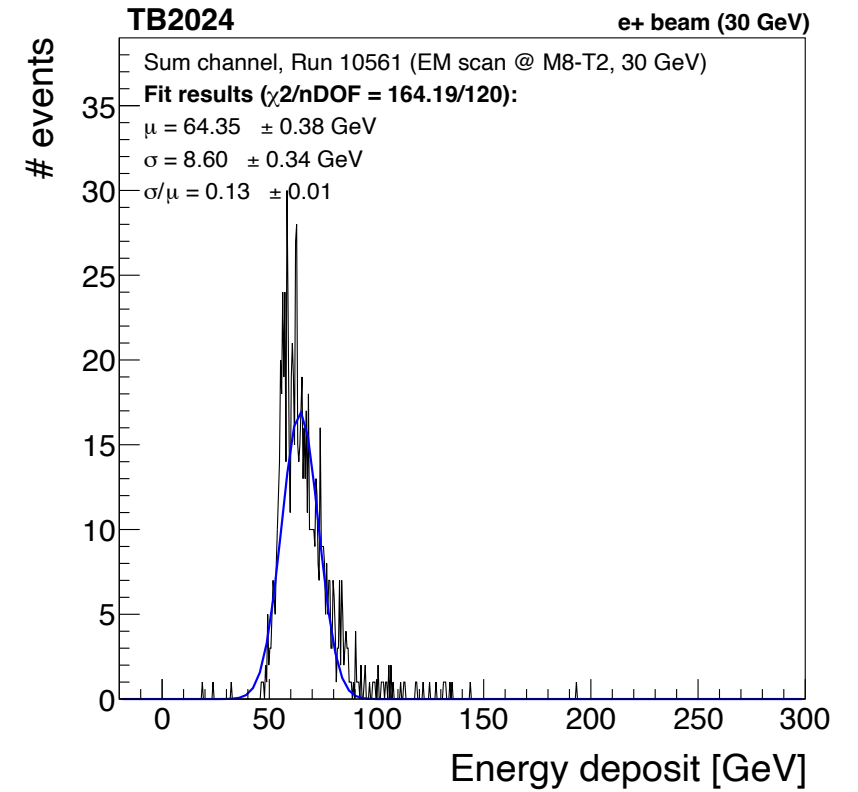
C channel



S channel

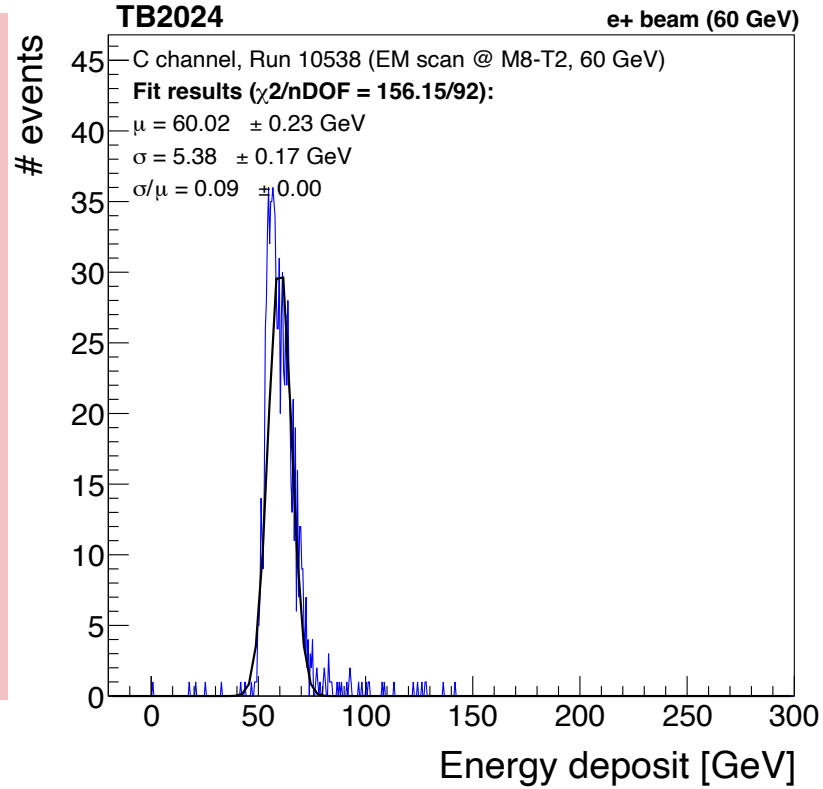


Sum (C+S)

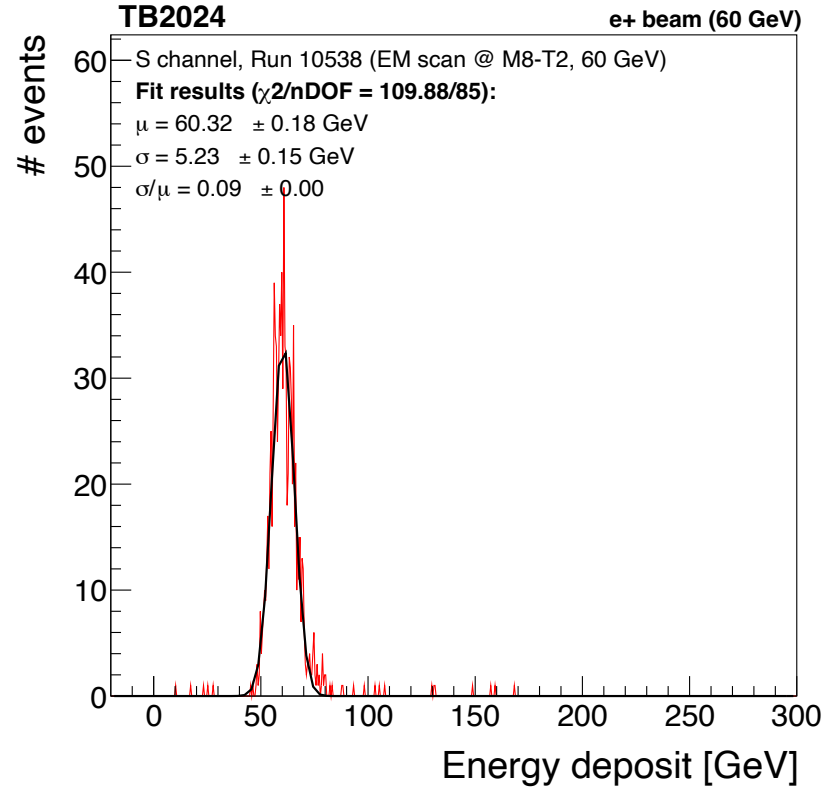


Energy Scan Results (Center: M8-T2)

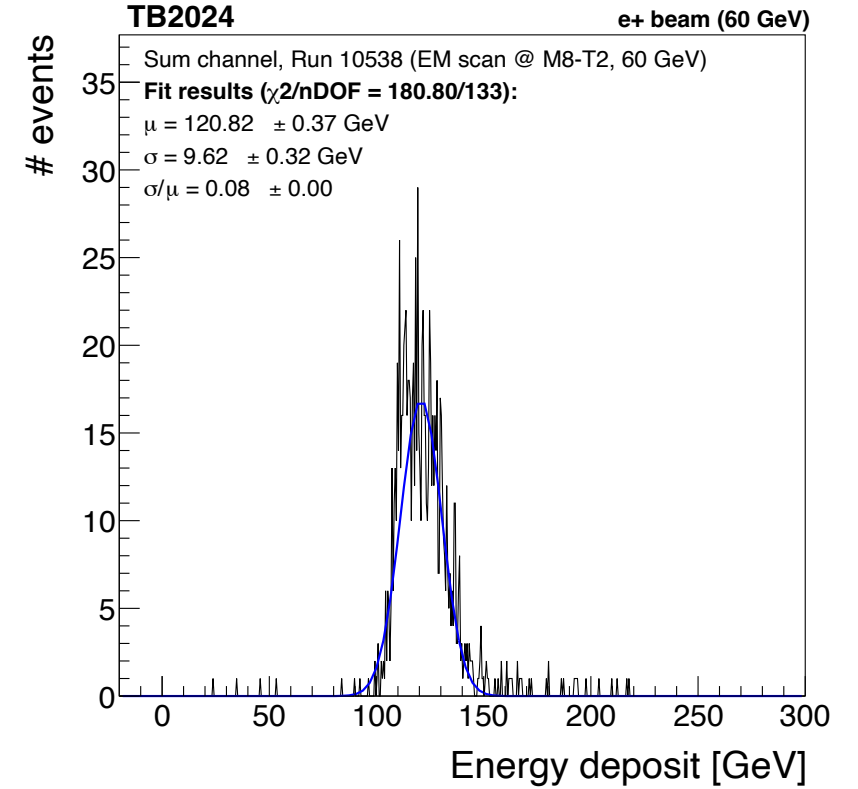
C channel



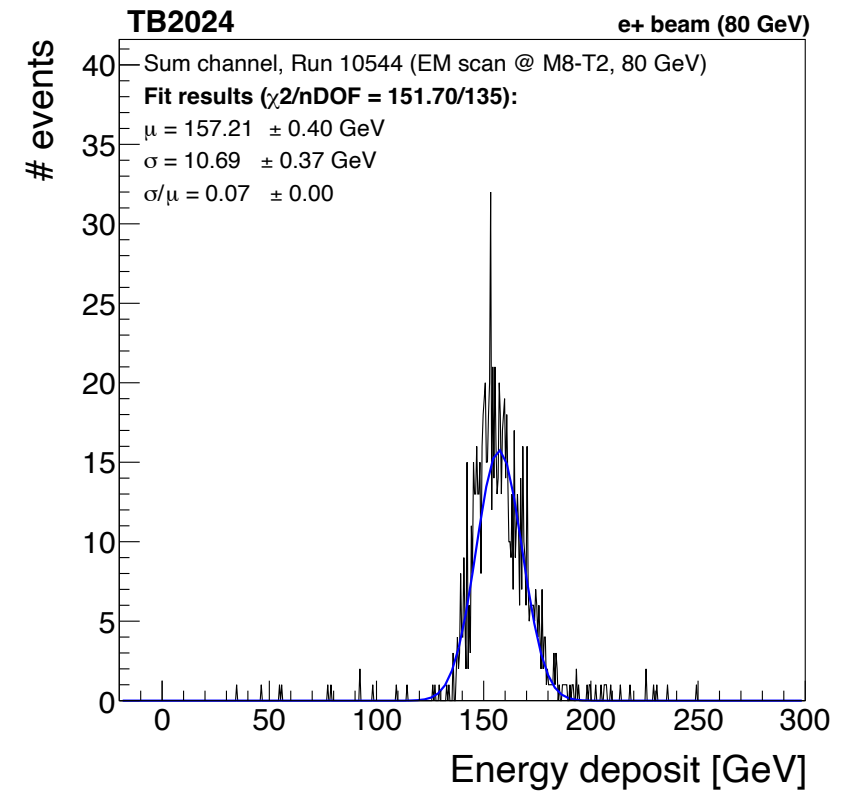
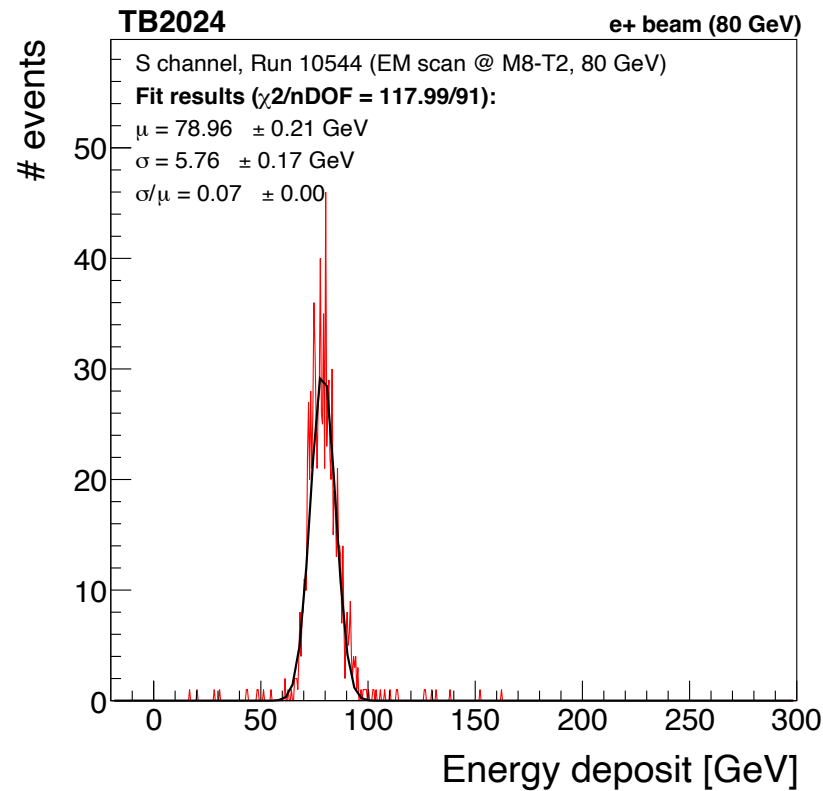
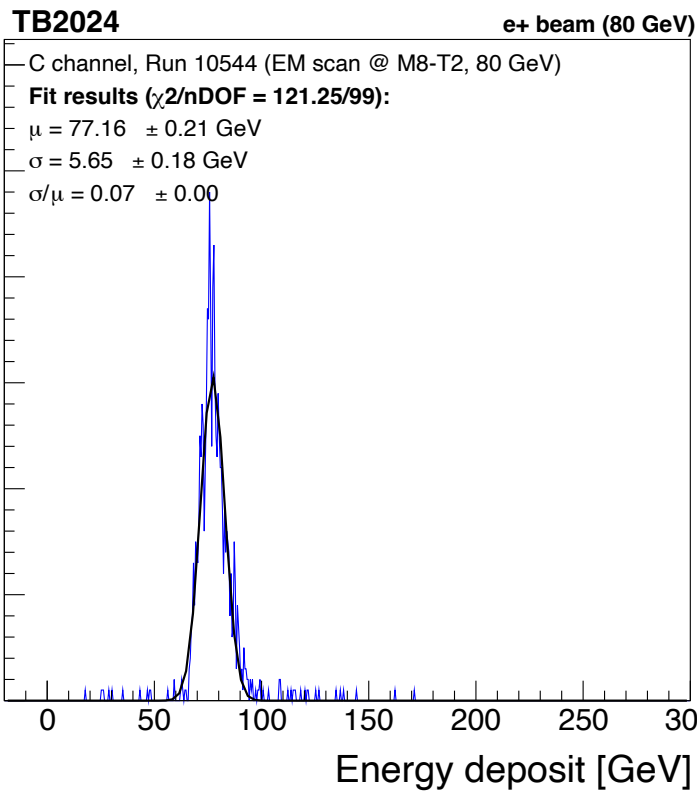
S channel



Sum (C+S)

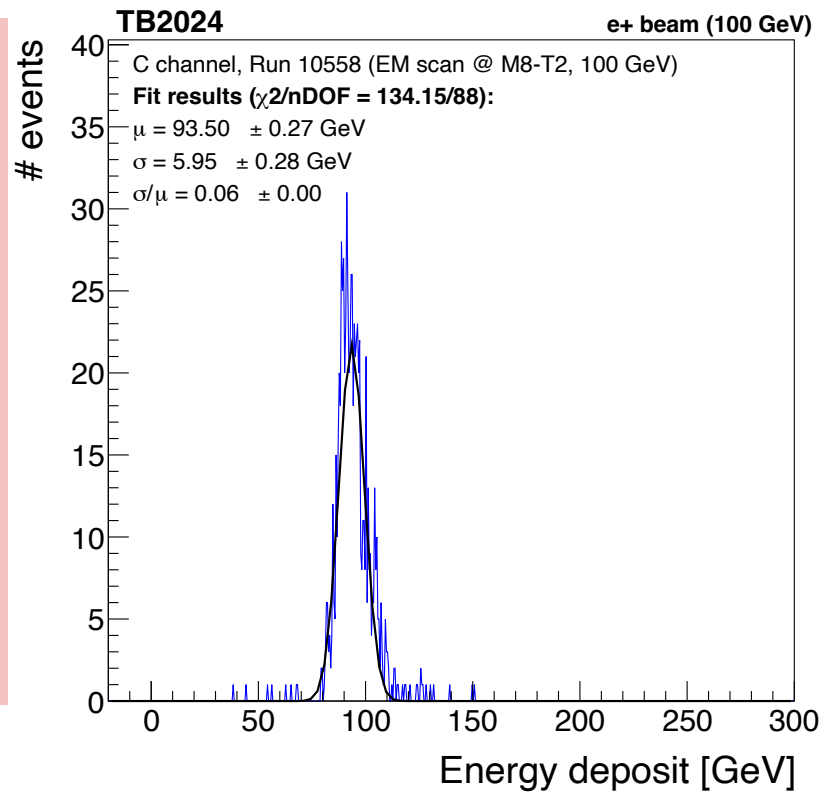


60 GeV

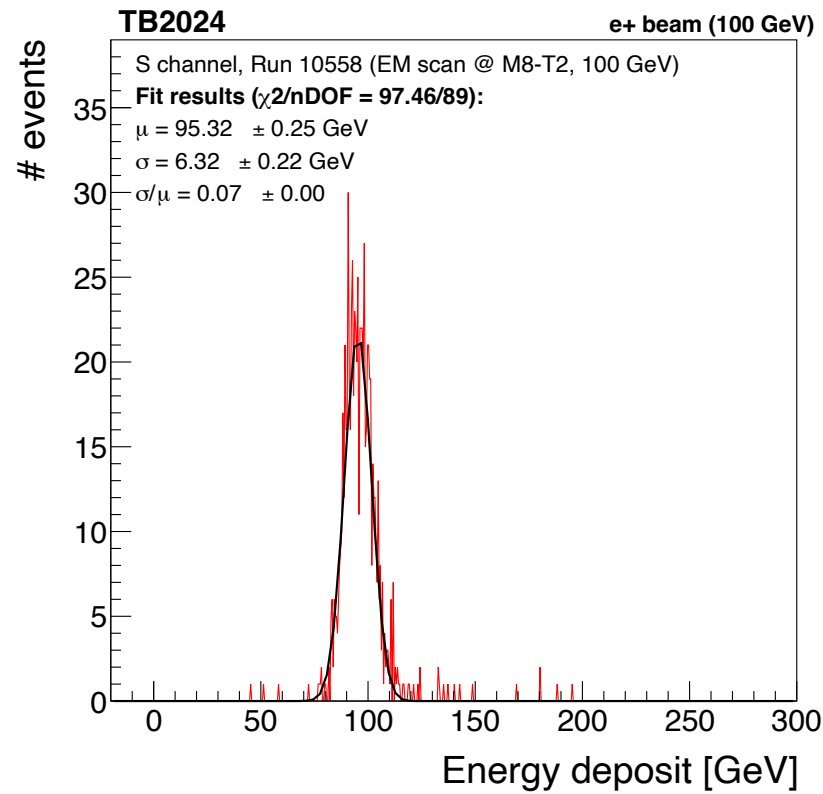


Energy Scan Results (Center: M8-T2)

C channel



S channel



Sum (C+S)

